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Prenom NOM 2 Affiliation	*Rapporteur*
Prenom NOM 3 Affiliation	*Rapporteur*
Prenom NOM 4 Affiliation	*Rapporteur*
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Remerciements

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Résumé

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Mots clés : mot clé 1, mot clé 2, mot clé 3, mot clé 4

Abstract

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Keywords : keyword 1, keyword 2, keyword 3, keyword 4

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Chapitre 1

Introduction

Objectifs

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1.1 Quelques conseils d'utilisation

1.1.1 Organisation du projet

Fichier maître à compiler Le fichier `main.tex` contient tous les sous-fichiers à inclure. C'est l'organisation du document.

Configuration Le dossier `config/` contient les fichiers de parametrage du document, les macros, ainsi que le package pour la couverture PSL (section 1.1.2).

Chapitres Ils sont situés dans des dossiers indépendants, avec des dossiers de figures associés (`.../fig/`). L'auto-complétion permet de trouver les figures associées au chapitre voulu : `ch1/fig/xxxx.jpeg`.

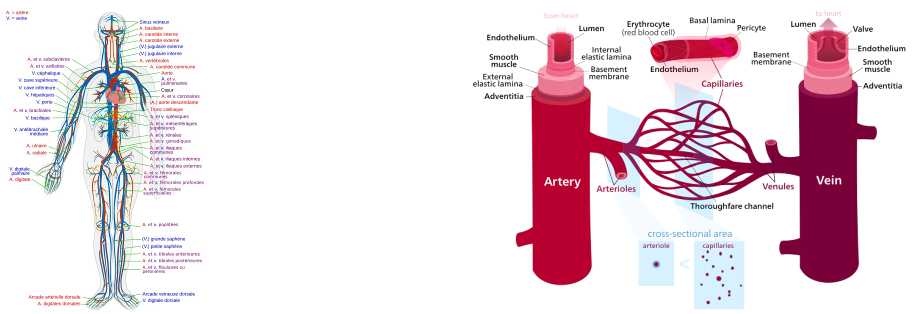
Autres fichiers et pages Les autres fichiers de texte (conclusion, remerciements), et information (info, liste des publications, glossaire) sont situés dans le dossier `others/`.

1.1.2 Couvertures par Pierre Guillou pour PSL

Seules les couvertures sont imposées par PSL. Le fichier ici a été réalisée par Pierre Guillou [1] et est disponible sur le site dans les ressources du site de PSL [2] (Version 1.2 - 20 juillet 2019). A noter que le président du jury doit être sur la première ligne de la liste du jury.

1.1.3 Illustrations

Overleaf propose l'auto-completion pour les figures de figures, ce qui est assez rapide et efficace. Un label est ajouté pour chaque figure afin de la citer après. La figure suivant est référencée par le label `fig1_cir` car c'est une figure du chapitre 1. Pour une table du chapitre 1, on pourra utiliser `tab1_power`. La figure [fig. 1.1a] illustre ce propos, pour plus de détails voir le chapitre 2.



(a) Réseau vasculaire

(b) Réseau capillaire sanguin

FIGURE 1.1 – (a) Le réseau vasculaire est composé des réseaux artériel et veineux. (b) Les capillaires sanguins réalisent la jonction entre les deux réseaux. [Wikimedia]

1. https://pierre.guillou.net/psl-cover/2018/
2. https://collegedoctoral.psl.eu/doctorat-psl/espace-ressources/

1.2 Une deuxième section

1.2.1 Acronymes

C'est assez pratique d'utiliser des acronymes, mais il faut les définir avant.

Ici, un commentaire

A faire

Ici on va utiliser le GPU ou la *Random Access Memory*, ou encore *Solid-State Drive* (SSD).

1.2.2 Quelques citations

1.2.2.1 Avec la commande Cite

On peut citer des références [**ueda_tissue_2020** ; **thomas_microbubble_2013**] pour appuyer le propos. La table des matières est réalisée avec `biblatex`.

1.2.2.2 En synchronisant Zotero

Zotero peut être synchronisé pour faire la gestion de la bibliographie. Il ne suffit que de mettre à jour le `.bib` de temps en temps.

Chapitre 2

CORPUS-CDI: comparing expressive vocabulary growth in children and language models

Objectifs

Introduce CORPUS-CDI, a count-based lexical benchmark aligning language model generation with child word production across developmental stages, and use it to compare how children and character-level LMs acquire vocabulary under matched input budgets.

2.1 Introduction

Large language models (LLMs) have shown that a general-purpose architecture trained on large text corpora can acquire rich syntactic and semantic knowledge [Brown, 2020 ; Devlin, 2019 ; Touvron, 2023]. Beyond their impact on NLP applications [Bommasani, 2021], LLMs provide scalable tools for testing theories of early language acquisition [Dupoux, 2018 ; Evanson, 2023 ; Lavechin, 2022 ; Apidianaki, 2024 ; Hu, 2025]. In particular, they offer a controlled way to investigate the long-standing hypothesis that children bootstrap into language learning by accumulating statistics from linguistic input [Saffran, 1996].

A central challenge when applying LLMs as models of the early language learner is establishing valid *quantitative* comparisons between the learning trajectories of real children and those of LLMs fed with comparable data. For instance, it has been claimed that LLMs require significantly more input data than children to reach the same level of performance [Frank, 2023]. But measuring this 'data gap' depends critically on developing sound metrics that can be applied consistently to both human learners and computational models.

We focus on measures of vocabulary, as there is a wealth of developmental data that provides human side of the comparison. Previous work has primarily explored two approaches for evaluating vocabulary in models: (i) **familiarity metrics**, which exploit LLMs' probabilistic outputs (e.g., surprisal or word–pseudoword preference) to probe lexical knowledge [Chang, 2022 ; Ma, 2025 ; Lavechin, 2025] ; and (ii) **cross-modal matching**, which assesses word–referent grounding using multimodal training and behavioral benchmarks [Vong, 2024 ; Tan, 2024 ; Nikolaus, 2021c], analogous to looking-while-listening paradigms in children [Swingley, 2011]. While informative, familiarity metrics often rely on limited behavioral data, and cross-modal methods require multimodal input unavailable to unimodal language models. Also, neither of them has focused on expressive vocabulary.

To overcome these limitations and enable large-scale, quantitative comparison with child vocabulary, we introduce CORPUS-CDI, a benchmark that aligns model generations with children's expressive vocabulary. CORPUS-CDI calibrates three key dimensions: (1) training exposure, scaled from 6 to 36 months based on daylong recordings [Cristia, 2019] ; (2) cumulative generated word count, matched to children's monthly word production rates; and (3) evaluation vocabulary, matched in part-of-speech categories and frequency distribution to the CDI word list [Frank, 2017].

Using CORPUS-CDI, we evaluate character-level LSTMs and Transformers trained on child-directed speech (CDS) and audiobook transcripts. We assess vocabulary acquisition across three complementary dimensions: lexical diversity (type-token ratio), word form accuracy (non-word rate), and vocabulary size (CORPUS-CDI score). Our analyses reveal that language models diverge from children in critical ways: models show excessive lexical diversity, minimal improvement in word form accuracy, and slower vocabulary growth. Frequency analyses further indicate that these gaps stem from long-tail truncation: low-frequency words are severely underrepresented in model acquisition, while high-frequency function words are overrepresented relative to children's productions.

2.2 Related Work

2.2.1 Developmental Trajectories

Prior work has examined the alignment between language model learning and child language development across multiple linguistic levels. A key challenge is mapping model training onto corresponding developmental milestones. Early studies used training checkpoints as proxies for age, tracking when models acquire words [Chang, 2022] or syntactic structures [Evanson, 2023]. However, these approaches repeatedly expose models to identical input batches, unlike children who encounter increasingly varied linguistic input. To address this, Lavechin et al. [Lavechin, 2023] varied input size to model developmental trajectories, improving zero-shot lexical accuracy with increased exposure. Beyond vocabulary, studies have investigated phonological development via word segmentation [Goriely, 2024], morphological acquisition through overregularization patterns [Haga, 2024], and syntactic learning [Evanson, 2023]. Most prior work, however, calibrates only input quantity, leaving output production and evaluation frequency uncontrolled. In contrast, our framework aligns training exposure, generation quantity, and evaluation timing to enable direct comparison between models' and children's expressive vocabulary.

2.2.2 Vocabulary Evaluation

Evaluating vocabulary acquisition in models has followed two main strategies: developing developmentally inspired metrics or applying existing metrics to child-relevant tasks [Kosoy, 2023 ; Nikolaus, 2021a ; Jiang, 2023]. Notably, Chang et al. [Chang, 2022] first introduced the MacArthur-Bates Communicative Development Inventory (CDI) to evaluate when models acquire individual words during training. Their findings highlight that models rely disproportionately on input frequency, with more variance in acquisition timing than child age-of-acquisition norms. While children also benefit from Zipfian distributions, the challenge for models is primarily learning words in the *long tail* of the distribution[Wolters, 2024 ; Lavi-Rotbain, 2023 ; Lavi-Rotbain, 2022]. This partially explains the gap between model and human acquisition patterns, particularly for low-frequency words.

Cross-linguistic analyses show that contextual surprisal predicts age of acquisition for function words and predicates, but not nouns [Portelance, 2023], suggesting distributional learning mechanisms affect lexical categories differently. To address this, our evaluation sets are frequency-matched *and* controlled for part-of-speech categories to ensure that differences are not confounded by lexical class or frequency effects.

2.2.3 Calibrated Evaluation Frameworks

Direct comparisons between models and children require alignment across training data, learning objectives, and evaluation tasks[Warstadt, 2022]. The BabyLM Challenge [Warstadt, 2023 ; Hu, 2024 ; Charpentier, 2025] established shared tasks by restricting model training to 10M or 100M words to approximate child exposure at ages 2 and 13. Despite these constraints, findings remain mixed: some studies report benefits from child-directed speech [Feng, 2024a], whereas others observe that Wikipedia training outperforms CHILDES-based input [Padovani, 2025b].

Input representation is another consideration: most models use BPE tokens [Huebner, 2021a ; Evanson, 2023], whereas infants perceive continuous acoustic speech [Dupoux, 2018]. We use character-level inputs, forcing models to learn word boundaries unsupervised from orthographic sequences [Bunzeck, 2024 ; Goriely, 2024]. This provides an upper bound on performance compared to phoneme-based approaches.

Finally, production metrics such as lexical diversity often use the type-token ratio (TTR), which is sample-size dependent [Richards, 1987] and requires fixed-size windows for comparability [McKee, 2022]. Sampling temperature introduces tradeoffs between diversity and quality, with no single setting optimizing both [Verine, 2025].

Our framework addresses these challenges through three calibrations: (1) age-appropriate training exposure derived from daylong recordings [Cristia, 2019], (2) production quantity matched to children's monthly output, and (3) frequency- and POS-matched evaluation sets reflecting child-directed speech patterns. Together, these enable a developmentally grounded evaluation of expressive vocabulary.

2.3 Methods

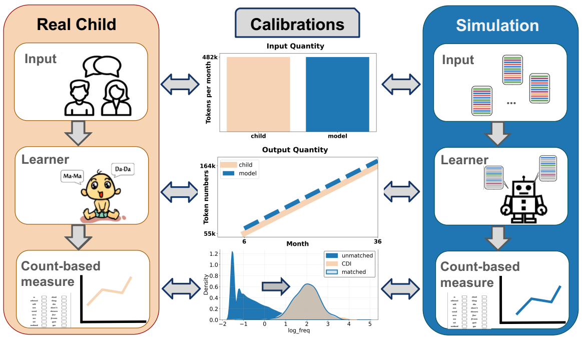


FIGURE 2.1 – **Overview of the corpus-CDI benchmark framework.** (a) Training and generation sizes are aligned with age-appropriate estimates of linguistic exposure and production from 6–36 months [Cristia, 2019]. (b) Evaluation sets match the frequency distribution of MacArthur–Bates CDI words in child-directed speech (English-CHILDES).

2.3.1 corpus-CDI Benchmark Overview

The CORPUS-CDI benchmark enables direct comparison between language model generation and child production through three methodological calibrations to address common mismatches in prior work (Figure 2.1):

(1) Age-appropriate training exposure: Training sets are scaled to match cumulative linguistic input from 6–36 months of age estimated from daylong recordings [Cristia, 2019].

(2) Production quantity matching: Generation sizes are calibrated to children's monthly word production rates estimated from longitudinal corpora [Cristia, 2019] to ensure comparable sample sizes for vocabulary assessment.
(3) Frequency-matched evaluation words: The evaluation vocabulary is drawn from the MacArthur–Bates Communicative Development Inventory [Fenson, 2007] and frequency matched to words in English CHILDES corpora.

We evaluate models on three dimensions: *lexical diversity* (type–token ratio), *word form accuracy* (non-word rate), and *expressive vocabulary size* (CORPUS-CDI score).

2.3.2 Training Data

We construct two training corpora representing different linguistic environments:
Audiobook corpus: English audiobook transcripts from LibriVox [Kearns, 2014], representing narrative-rich, syntactically complex adult-directed language. **Child-directed speech (CDS) corpus:** English CHILDES transcripts [MacWhinney, 2000], representing the naturalistic register of early child input.

Age-appropriate scaling Following CRISTIA et al. [Cristia, 2019], we estimate exposure from 250 hours at 6 months to 1,500 hours at 36 months of age. This corresponds to approximately 500 hours of speech input per year, representing a mid-range estimate of children's linguistic environments. Converting speech hours to words using a standard rate of 3 words per second yields training sets ranging from 2.89M words (6 months) to 17.35M words (36 months). Developmental datasets are constructed every 6 months by randomly sampling the target quantity from each corpus (see Table 2.1). Given English CHILDES contains a total of 15.6M words of caregiver inputs, the available data is equivalent to those at the 30-month level, which thus serves as the maximum training input. We emphasize that this scaling controls for *total input quantity* rather than assuming uniform structure of children's linguistic environments across months.

Preprocessing All text is lowercased, with spaces and basic punctuation (periods, commas, apostrophes) preserved as word boundary markers. No tokenization or subword segmentation is applied, forcing models to infer word boundaries directly from raw character sequences.

2.3.3 Model Architectures and Training

We train character-level autoregressive models under two architectures: **LSTM** Three layers with 200-dimensional character embeddings, 1024-dimensional hidden states, and 200-dimensional output projections, following LAVECHIN et al. [Lavechin, 2023]. **Transformer** GPT-2-style decoder-only architecture with 12 transformer layers, 512-dimensional hidden states, 12 attention heads per layer, following GORIELY et al. [Goriely, 2024].

Training procedure Models are trained to minimize cross-entropy loss using the Adam optimizer (learning rate 0.0001, $\beta_1 = 0.9$, $\beta_2 = 0.98$) with inverse square root learning rate decay and 10,000 warmup steps. We train until validation perplexity converges. For each age-corpus pair, two models are trained with same random seeds and non-overlapping data subsets. Training details are provided in Table 2.2.

Generation procedure At inference time, we generate text using temperature sampling with $T \in \{0.3, 0.6, 1.0, 1.5\}$. Generation begins with a beginning-of-sequence (BOS) token and continues until reaching the target word count corresponding to children's estimated monthly production at that developmental stage (see Table 2.3 in Appendix). Character sequences are segmented into words using space and punctuation boundaries. For months 31–36, outputs are generated from the same model checkpoints. Because the CDS corpus contains a total of 15.6M words of caregiver input, this limits the age-appropriate training to the 30-month equivalent. For months 31–36, we generate outputs using the same 30-month checkpoint, scaling generation lengths according to children's estimated production, while noting that the training exposure does not increase beyond 30 months. This ensures a consistent comparison across developmental stages while respecting the corpus limitations.

2.3.4 Evaluation Set Construction

Base vocabulary and POS filtering We start with the MacArthur-Bates Communicative Development Inventory (CDI) [Fenson, 2007], a standardized checklist of 680 English words with normative age-of-production data. To control for lexical class effects, we apply POS tagging using a RoBERTa-based SpaCy pipeline. For each word, POS labels are aggregated across contexts and majority voting is applied separately in each corpus. Function words (articles, pronouns, prepositions, auxiliaries) are excluded, retaining nouns, verbs, adjectives, and adverbs.

Frequency stratification Filtered CDI words are divided into six frequency bands based on their occurrence in child-directed speech (adult input in English CHILDES), with an equal number of words per band.

Audiobook subset selection For a comparable evaluation, we construct a matched audiobook subset using the same POS filtering and frequency banding. Words are selected such that their frequency distribution mirrors the filtered CDI evaluation set. This process yields two aligned datasets: i.) Filtered CDI evaluation set with 420 words in six frequency bands; and ii.) frequency-matched audiobook subset with the same size per frequency band. This enables a developmentally grounded comparison between model generations and children's expressive vocabulary.

2.3.5 Evaluation Metrics

We assess model performance using three complementary metrics capturing lexical diversity, form accuracy, and expressive vocabulary.

Type-Token Ratio (TTR) Measures lexical diversity as the ratio of unique word types to total word tokens. Following Räsänen et al. [Räsänen, 2024], we compute TTR over fixed-size chunks of 3,500 words to control for sample size effects, then average across all chunks.

Non-word Rate Quantifies word form accuracy as the proportion of generated word types not found in comprehensive English dictionaries: CELEX [Van Heuven, 2014], the Enchant library, [1] and Wiktionary. [2] A word is classified as valid if it appears in any of these three resources. This metric captures models' ability to produce orthographically valid English words.

corpus-CDI Score Adapts the human CDI vocabulary assessment to model generations through a count-based criterion. We mark a word as "known" if it appears at least τ times in the generated sample. To determine τ, we analyze CHILDES child productions and fit the resulting vocabulary growth curve to the empirical CDI trajectory. We select $\tau = 60$, which yields slopes most consistent with observed developmental patterns. The CORPUS-CDI score is computed as:

$$\mathrm{CDI} = \frac{|\{w \in \text{eval set} : \text{count}(w) \geq \tau\}|}{|\text{eval set}|}$$

where the evaluation set includes 420 words. This value represents the proportion of age-appropriate vocabulary that the model generates with sufficient frequency to be considered "known." Sensitivity analyses for alternative thresholds ($\tau \in \{1, 500\}$) are shown in Figure 2.6.

2.4 Results

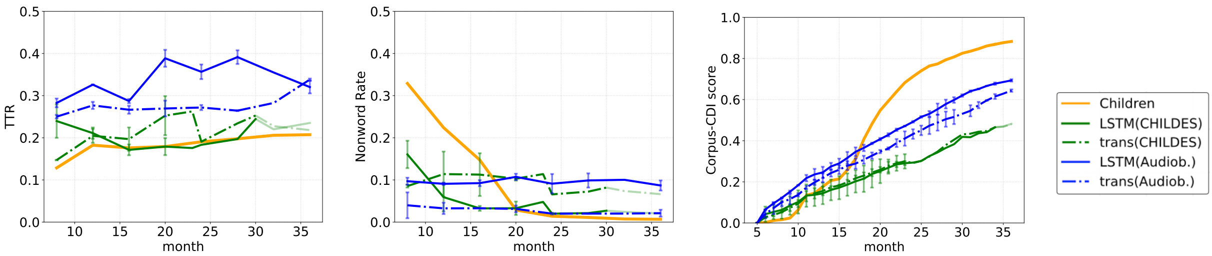


FIGURE 2.2 – **Core metrics comparing child and model vocabulary development across age groups.** Left: Type–token ratio (TTR) measuring lexical diversity. Middle: Non-word type rate indicating the proportion of invalid word forms. Right: CORPUS-CDI scores tracking vocabulary acquisition (words counted as known if produced >60 times). Results are shown for LSTM and Transformer models trained on audiobooks and child-directed speech (CDS), compared with CHILDES child production data. Error bars denote 95% confidence intervals across disjoint training subsets. For the final months (32–36), outputs are generated from the same model checkpoint and shown with lighter transparency.

2.4.1 Developmental Trajectories

Figure 2.2 presents model performance across three complementary metrics as a function of cumulative training exposure (6-36 months), compared against children's developmental trajectories from CHILDES production data. All results use temperature $T = 1.0$ as the default setting.

1. https://pypi.org/project/pyenchant/
2. https://www.wiktionary.org

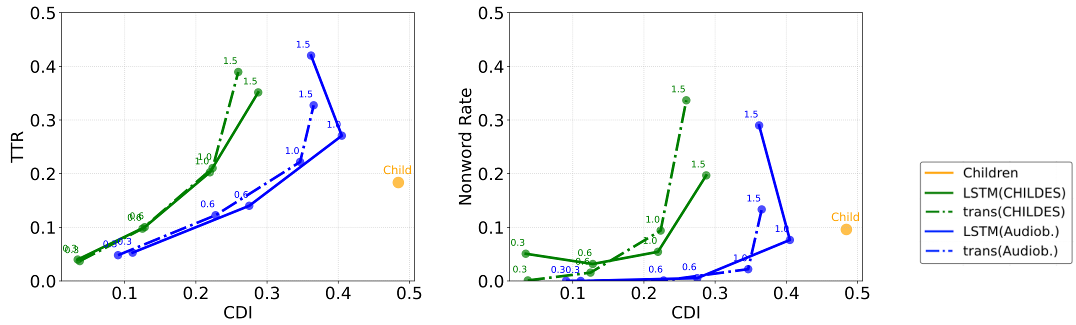


FIGURE 2.3 – **Temperature trade-off across different metrics; temperature settings (0.3-1.5)** Left: Trade-off between type-token ratio and CDI score. Right: Relationship between non-word rate and CDI score. Points show mean scores across models, with numbers illustrating temperature values.

Lexical diversity Both LSTM and Transformer models exhibit higher type-token ratios (TTR) than children across developmental stages. Models trained on audiobooks show slightly higher TTR than those trained on child-directed speech, reflecting broader lexical variety in narrative text. Unlike children, whose TTR gradually increases with age, model TTR remains relatively stable. This suggests models generate more lexically diverse outputs from early on, but does not increase substantially with additional training data.

Word form accuracy Models and children show different developmental patterns in non-word production rates. Children begin with high non-word rates that decrease sharply by the 24th month, reflecting the gradual development of articulatory control and phonological mastery. In contrast, both model architectures maintain relatively stable non-word rates across all training quantities, showing minimal improvement with increased data exposure. Transformers produce slightly fewer non-words than LSTMs, but neither captures the steep improvement observed in children. This divergence likely reflects a critical limitation of text-based models: they cannot capture the development of articulatory-motor control that drives children's decreasing error rates.

Vocabulary growth CORPUS-CDI scores show that children's vocabulary grows sigmoidally, with rapid acceleration during the vocabulary spurt. Models, by contrast, display roughly linear growth, with Transformers slightly outperforming LSTMs and audiobook-trained models outperforming CDS-trained ones. Critically, neither architecture reproduces the accelerating trajectory characteristic of human vocabulary development even as training data accumulates as the estimated quantity.

Across all three metrics, models exhibit qualitatively different developmental patterns from children. They show excessive lexical diversity, minimal improvement in word form accuracy, and slower vocabulary growth that lacks the characteristic acceleration of human development. These differences persist across both architectures and training corpora, suggesting they reflect fundamental limitations in how current language models learn from sequential exposure rather than artifacts of specific model choices. In the following sections, we investigate two potential explanations for these gaps: whether alternative temperature settings during decoding can better align model behavior with children (Section 2.4.2), and whether the observed differences relate to models' frequency-dependent learning patterns (Section 2.4.3).

2.4.2 Temperature Tradeoff

The developmental gaps observed in Section 2.4.1 raise an important question: could these differences arise from suboptimal generation parameters rather than inherent learning limitations? Temperature sampling controls generation stochasticity, with lower temperatures favoring high-probability outputs and higher temperatures increasing diversity. We evaluate whether any temperature setting can align models with children across all three metrics.

Figure 2.3 shows the relationships between metrics as temperature varies from 0.3 to 1.5. Each point represents the mean performance across all developmental stages for a given architecture, corpus, and temperature combination.

TTR-CDI trade-off Models exhibit a clear trade-off: increasing temperature simultaneously raises type-token ratio (TTR) and CORPUS-CDI scores. Low temperatures produce TTR values closer to children's range but severely underrepresent vocabulary size, whereas high temperatures improve vocabulary coverage at the cost of excessive lexical diversity. No temperature setting positions models within the child-like range for both metrics. Audiobook-trained models consistently overshoot children's TTR, while CDS-trained models come closer but still fall short in vocabulary size. This indicates that the repetitive, high-frequency vocabulary characteristic of early child speech cannot be recovered through decoding adjustments alone.

Non-word-CDI trade-off Higher temperatures increase non-word rates for both architectures. While some CDS-trained models at intermediate temperatures approximate children's average non-word rates, this alignment is misleading: children's non-word rates decrease sharply over development, whereas models remain largely static. Moreover, achieving this partial alignment comes at the expense of TTR matching, suggesting that it is still implausible to simultaneously satisfy multiple developmental constraints.

These results demonstrate that temperature tuning cannot resolve the developmental misalignment between models and children. The gaps reflect fundamental differences in how models and children balance vocabulary size, lexical diversity, and production accuracy. Post-hoc decoding adjustments cannot substitute for differences in underlying learning and production mechanisms. In the next section, we examine whether these differences relate to frequency-dependent learning patterns.

2.4.3 Frequency effect

The analyses so far show that models lag far behind children in vocabulary acquisition, and this gap cannot be closed by tuning generation parameters. We now ask whether this delay is uniform across input frequency. To test this, we divided the vocabulary into six frequency bands based on child-directed speech and tracked the proportion of words acquired in each band, using

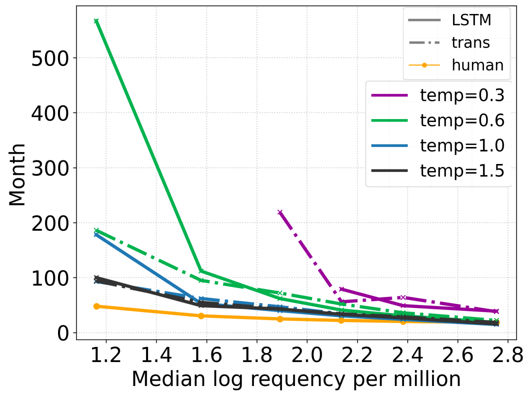


FIGURE 2.4 – **Extrapolated months across different datasets for models trained on audiobooks**. x-axis: log frequency per million in training data; y-axis:estimated developmental month at which models reach 80% mastery for each frequency band. Missing points of the lower frequency bands of temperature =0.3 come from 0

the same sigmoid growth curve fitting as above. For each band, we extracted the *extrapolated age of acquisition*, defined as the cumulative training exposure at which the model reaches 80% mastery, matching the maximum rate observed in CDI norms. This "months-to-mastery" measure quantifies how quickly different frequency bands are learned as a function of cumulative training exposure.

Divergent frequency sensitivity Figure 2.4 shows that children show much weaker frequency dependence than models, with only modest variation between high- and low-frequency vocabulary. By contrast, both models display considerable frequency dependence: high-frequency words are acquired relatively quickly, but acquisition times for low-frequency words increase by orders of magnitude. This indicates that models rely heavily on repeated exposure, whereas children generalize more effectively from sparse input.

Temperature modulation Temperature settings systematically modulate frequency effects. At low temperatures ($T = 0.3$), models fail to acquire rare words, as deterministic decoding prevents them from being generated sufficient times to count as "known". Higher temperatures ($T = 1.0$–1.5) alleviate this issue by surfacing rare words more often, but the improvement remains limited: even under the most favorable conditions, models require many times more exposure than children to acquire low-frequency vocabulary. Thus, temperature affects severity of the bias but not the underlying cause.

Long-tail truncation These results point to long-tail truncation as the primary mechanism underlying models' vocabulary acquisition gap. Language models are trained to minimize cross-entropy loss, which optimizes for predicting the most probable next character given context. This objective inherently prioritizes frequent patterns: prediction errors on common words contribute more to loss than on rare words, leading models to allocate representational capacity proportionally to frequency.

Children, by contrast, do not optimize for predictive accuracy. Their learning mechanisms such as explicit memory for individual word-learning episodes [Markson, 1997], cross-situational inference[Smith, 2008], and social-pragmatic cues [Tomasello, 2000] allow them to acquire words from sparse observations without requiring proportional exposure. This qualitative difference in learning objective, rather than mere data inefficiency, explains why children succeed in the long tail where models consistently fail.

2.4.4 Production of Content Words

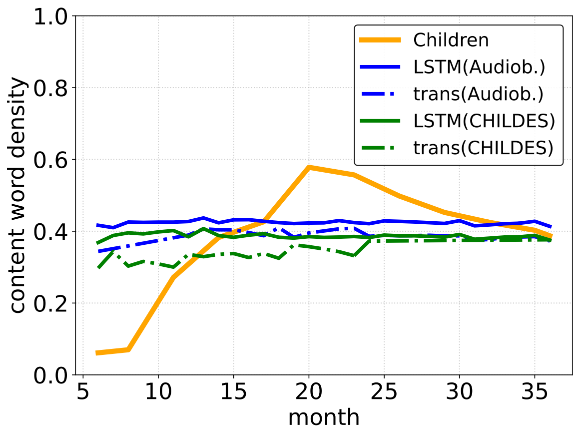


FIGURE 2.5 – **Proportion of content words in generated text across development.** Children show an inverse U-shaped trajectory, while all model configurations maintain stable proportions around 40% across training stages.

The frequency-dependent learning patterns in Section 2.4.3 may partly stem from differences in functional versus content word production. Function words dominate the highest-frequency bands [Zipf, 2016], whereas our CORPUS-CDI evaluation targets content words (Section 2.3.4). Over-reliance on function words could explain why models achieve lower CDI scores and limited lexical diversity: frequent repetition of a small set of functional items increases token counts without expanding the number of unique word types.

Children's inverse U-shaped trajectory Figure 2.5 shows that children exhibit a pronounced inverse U-shaped trajectory in content word production. Early productions contain few content words, followed by a sharp rise during the vocabulary spurt, and then a decline as grammatical function words become integrated [Bates, 1994; Brown, 1973]. Critically, this declining phase coincides with children's steepest CORPUS-CDI growth (Figure 2.2), showing that children's vocabulary growth is not simply driven by frequency exposure but by a qualitative reorganization of their lexical production.

Models' invariant balance In contrast, all model configurations maintain stable content-word proportions around 40%, which closely matches adult English distributions [Biber, 2021]. This invariance holds across architectures and training corpora: both CDS- and audiobook-trained models converge almost immediately to corpus-level statistics and remain fixed throughout training.

This invariance helps explain several earlier findings: (1) Frequency truncation (Section 2.4.3) disproportionately harms content words, which spread more evenly across frequency bands. (2) CDI underperformance reflects systematic undergeneration of content words, on which CDI norms are based. (3) Elevated TTR may paradoxically arise from over-allocating tokens to a small set of high-frequency function words, forcing broader but shallower sampling of content vocabulary.

Taken together, these results reveal a fundamental mismatch: cross-entropy training aligns models with corpus-level statistics but not with children's developmental trajectory. While children dynamically restructure their lexicon and functional–content balance, models remain tied to static frequency distributions.

2.5 Discussion

We introduced CORPUS-CDI, a benchmark that enables direct comparison between children's expressive vocabulary and language model generations by aligning training exposure, production quantity, and frequency distributions. Our analyses yield three main findings that highlight key differences between human and model vocabulary acquisition.

First, while both character-level LSTMs and Transformers expand their vocabularies with increased training data, their developmental trajectories diverge largely from those of children. Models consistently produce far greater lexical diversity, maintain stable non-word rates across training, and exhibit slower vocabulary growth without the characteristic acceleration observed in children. These patterns generalize across architectures (LSTM vs. Transformer) and training corpora (audiobooks vs. child-directed speech), suggesting that the underlying learning mechanisms driving word acquisition differ fundamentally between children and current language models. In other words, the mismatch is not tied to the choice of corpus or model architecture, but reflects mechanistic differences in how children versus models acquire and prioritize vocabulary.

Second, these divergences cannot be reconciled through parameter tuning at generation time. Temperature variation experiments show that no single setting simultaneously aligns models with children across TTR, non-word rate, and vocabulary size. The gap thus arises from fundamental properties of the learning and production mechanisms rather than from suboptimal decoding choices.

Third, frequency analyses reveal the mechanistic source of these differences. Children acquire words across frequency bands within a narrow age window (around 35–40 months), whereas models require exponentially more exposure for low-frequency words (equivalent to 300–600 months of input for words around log frequency 1.2–1.4 per million). This extreme frequency sensitivity follows directly from cross-entropy optimization, which down-weights rare words during training. Importantly, the interaction with word categories amplifies the divergence: models preserve adult-like content/function ratios (around 40% content words) throughout training, while children show a U-shaped trajectory driven by the integration of function words. Since function words cluster in high-frequency bands whereas content words (the CDI evaluation focus) distribute more evenly, models' frequency-proportional learning underproduces the vocabulary class children prioritize.

These findings point to limitations of predictive accuracy as the sole training objective for modeling acquisition. Children do not rely solely on distributional statistics; they integrate referential grounding, social-pragmatic inference, and episodic memory of word-learning events to acquire vocabulary from sparse observations. Progress in cognitively grounded language modeling may therefore require learning objectives beyond predictive accuracy, such as memory-augmented architectures for rare word retention, contrastive objectives that emphasize lexical representations, or interactive learning paradigms where production influences subsequent input.

2.6 Limitations

Our evaluation framework has several limitations. First, we evaluate character-level models rather than speech-based systems. While character input provides an upper bound by preserving invariant word forms and explicit boundaries, it bypasses the acoustic variability and segmentation challenges that infants solve [Dupoux, 2018]. Speech-based language models [Lakhotia, 2021 ; Nguyen, 2024] must additionally contend with phonetic variability and ambiguous word boundaries. As such, our reported gaps likely underestimate the difficulty of vocabulary acquisition in more realistic, speech-based models. Extending CORPUS-CDI to speech therefore requires addressing the non-trivial task of mapping continuous acoustic output into discrete word forms suitable for CDI scoring and we would expect even larger divergences from children's developmental trajectories. Moreover, our models lack multimodal grounding—the word-object associations and interactive feedback that children exploit. Such grounding likely supports children's frequency-independent learning by providing alternative cues beyond distributional statistics.

Second, the CHILDES transcripts used as reference data introduce possible biases. Transcribers often normalize children's productions by correcting errors or completing fragments, which may underestimate non-word rates and inflate apparent model–child gaps. Conversely, our character-level models inherently generate orthographically valid forms, potentially overstating their production accuracy relative to children's speech.

Third, our comparison of parental CDI reports with model outputs involves methodological misalignment. CDI scores reflect whether parents judge a child to "say" a word, capturing semantic knowledge and contextual usage in addition to production capability. In contrast, our CORPUS-CDI metric counts raw production frequency in unconstrained model text. A child uttering dog once demonstrates robust lexical knowledge, whereas a model generating dog 60 times may not. While our threshold ($\tau = 60$) calibrates vocabulary growth slopes, the construct validity gap between CDI reports and model generation remains.

Finally, our study focuses on English and a single vocabulary assessment tool (CDI). Cross-linguistic validation is essential for establishing universality: languages with richer morphology, less transparent orthography, or different child-directed registers may yield different alignments between model and child learning. Character-level models in particular may be sensitive to orthographic properties, limiting generalizability beyond English.

2.7 Ethics Statement

All data used in this study are publicly available through the CHILDES database. Our work focuses exclusively on model benchmarking and internal simulations, with the goal of advancing fundamental research on language acquisition. We emphasize that the evaluations conducted here are not sufficient for deployment in real-world applications. Any future use involving human data should undergo rigorous validation and appropriate de-identification procedures to ensure privacy and ethical compliance.

2.8 Appendix

2.8.1 Training and generation details

Table 2.1 summarizes the age-appropriate scaling of training data across three exposure scenarios. Linguistic exposure estimates are based on daylong recording studies [Cristia, 2019], which document substantial variability in the amount of speech heard by children across environments. Annual exposure ranges from approximately 100 hours in low-input households to 1000 hours in high-input ones, with 500 hours per year representing a mid-range estimate.

Cumulative exposure thus increases from 50–500 hours at 6 months to 300–3000 hours by 36 months. Assuming a standard speech rate of three words per second, this corresponds to training sets ranging from 0.58M words (6 months, low exposure) to 34.71M words (36 months, high exposure).

Age	**Hours**	**Words**	**Characters**	**Utterances**
Low exposure scenario (100 hours/year)				
6	50	0.58	2.95	58
12	100	1.16	5.89	116
18	150	1.73	8.84	173
24	200	2.31	11.78	231
30	250	2.89	14.73	289
36	300	3.47	17.67	347
Medium exposure scenario (500 hours/year)				
6	250	2.89	14.73	289
12	500	5.78	29.45	578
18	750	8.68	44.18	868
24	1000	11.57	58.90	1157
30	1250	14.46	73.63	1446
36	1500	17.35	88.35	1735
High exposure scenario (1000 hours/year)				
6	500	5.78	29.45	578
12	1000	11.57	58.90	1157
18	1500	17.35	88.35	1735
24	2000	23.14	117.80	2314
30	2500	28.92	147.25	2892
36	3000	34.71	176.70	3471

Table 2.1 – **Age-appropriate scaling of training datasets across three exposure scenarios.** Estimates are derived from three exposure scenarios (100h, 500h, 1000h per year), reflecting observed variability in children's linguistic environments [Cristia, 2019]. Word counts assume 3 words/second; character counts assume 5.1 characters/word; utterance counts assume 10 words/utterance.

Model training employed carefully tuned hyperparameters (Table 2.2), selected through systematic validation experiments to ensure training stability and computational efficiency across different data scales.

The examples in Table 2.4 highlight distinct generation behaviors across datasets and model

Hyperparameter	**Value**
Max sequence length	1024
Batch size	64
Learning rate	0.0001
Learning rate scheduler	inverse sqrt
Warmup steps	10000
Optimizer	Adam
Adam-beta1	0.9
Adam-beta2	0.98
Dropout	0.1
LSTM hyperparameter	**Value**
decoder layers	3
hidden size	1024
embedding dimension	200
Transformer hyperparameter	**Value**
Transformer layers	12
Intermediate hidden size	512
Attention heads	12
Attention dropout	0.05

Table 2.2 – **Language model training hyperparameters.** Configurations balance model capacity, stability, and efficiency across training scales.

architectures under a fixed sampling temperature ($T = 1.0$). Models trained on child-directed input (Child, CHILDES) tend to produce shorter, phonologically simple, and semantically grounded utterances, resembling early stages of language development. In contrast, audiobook-trained models generate longer, syntactically structured sequences but often lack pragmatic coherence. Comparing architectures, Transformers exhibit more globally consistent syntax and longer dependency spans, while LSTMs display stronger local predictability but higher rates of morphological irregularities. These qualitative patterns align with quantitative trends observed in our lexical and nonword diversity metrics, underscoring how data type and inductive bias jointly shape emergent linguistic structure.

2.8.2 Result details

To establish reliable vocabulary acquisition metrics, we conducted systematic analyses of different word count thresholds (Figure 2.6). This calibration process aimed to align our corpus-CDI scores with empirical observations of child vocabulary development.

Notably, the frequency effects observed in the Machine CDI are evaluated on a selected subset matched with the human-CDI set. It is unclear whether this effect is influenced by the randomness of sampling. Also, the selected words are typically in the median frequency bands, making it unknown whether such an effect stems from the limited generation size. To address this concern, we expanded our analysis of the model generations using the same and larger sampling sets. Figure 2.7 shows the word frequency distributions of the training and generated datasets by the LSTM model trained on 3.4M words, with different generation sizes. Expanding the

month	#sec/hour	estimation	production
6	61.25	55,125	782
7	65.30	58,769	1,323
8	69.35	62,412	1,278
9	73.40	66,056	3,964
10	77.44	69,699	627
11	81.49	73,343	1,148
12	85.54	76,986	1,361
13	89.59	80,630	568
14	93.64	84,273	2,720
15	97.69	87,917	13,765
16	101.73	91,560	3,620
17	105.78	95,204	7,589
18	109.83	98,847	15,197
19	113.88	102,491	17,090
20	117.93	106,134	19,912
21	121.98	109,778	33,458
22	126.02	113,421	33,880
23	130.07	117,065	63,738
24	134.12	120,708	191,062
25	138.17	124,352	130,525
26	142.22	127,995	136,195
27	146.27	131,639	148,741
28	150.31	135,282	158,910
29	154.36	138,926	190,263
30	158.41	142,569	195,549
31	162.46	146,213	196,677
32	166.51	149,856	184,024
33	170.56	153,500	185,534
34	174.60	157,143	185,691
35	178.65	160,787	153,515
36	182.70	164,430	299,537

TABLE 2.3 – **Monthly progression of speech metrics.** Estimated exposure and production rates over developmental time.

generation set mitigates some truncation effects, as more high-frequency words are generated, but low-frequency words remain underrepresented. This pattern reflects the model's limited ability to reproduce realistic word distributions, especially for rare words, even with larger sampling sizes.

Although models trained on higher-exposure estimates (1000 h) achieve slightly higher CORPUS-CDI scores than those trained on smaller sets (100 h, 500 h), the overall developmental trajectories remain similar. This pattern indicates that scaling input exposure alone does not yield more human-like developmental dynamics; instead, the gap likely arises from structural differences in learning mechanisms rather than data quantity.

Setting	Generated Sequences
Child	dad dada mommy looked it missed its wheel i want my plate for it
Audiobook (LSTM)	wounded for woman says no i don't know what i fuller over his week scully a
Audiobook (Trans)	to o he had been and the work of the lord ascend mass of softly rounded head
CHILDES (LSTM)	can bump he said I was wondering what yeah detfive wroggy on drh aulf
CHILDES (Trans)	on helsewhere he doesn't like him does he hafta dragged on doh aunty quite

TABLE 2.4 – **Examples of generated sequences.** Samples are drawn from models trained on different datasets and architectures at a sampling temperature of 1.0. The boundary marker is replaced with blank space for readability.

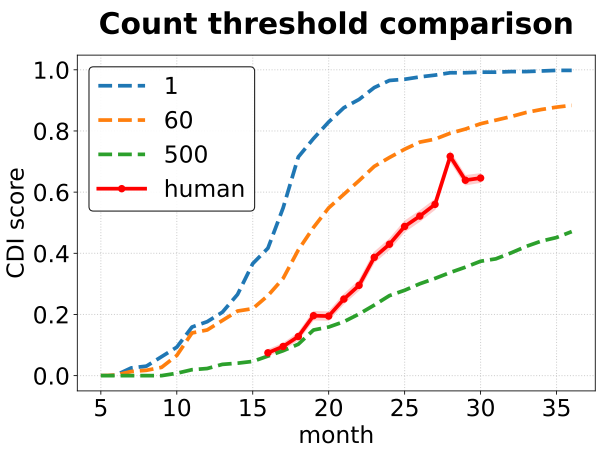


FIGURE 2.6 – **Count-based calibration** Comparison of different word frequency thresholds for determining word acquisition. The optimal threshold (60 occurrences) was selected based on alignment with parental vocabulary size reports (red line). Each threshold produces distinct vocabulary growth trajectories, with higher thresholds resulting in more conservative estimates.

Dataset (Model)	Temp	TTR (%)		NWType (%)		NWToken (%)		CDI(%)	
		Mean	Last	Mean	Last	Mean	Last	Mean	Last
CHILDES	-	18.3	20.7	9.6	0.7	4.7	0.2	48.6	88.4
Audiobook (LSTM)	**0.3**	43.8 [41.3, 46.1]	44.2	13.5 [12.0, 14.8]	13.8	10.4 [9.2, 11.6]	10.6	62.5 [59.2, 65.8]	63.1
	0.6	45.9 [43.4, 48.2]	46.3	14.8 [13.3, 16.1]	15.1	11.7 [10.5, 12.9]	11.9	65.2 [61.9, 68.5]	65.8
	1.0	48.1 [45.6, 50.4]	48.5	16.4 [14.9, 17.7]	16.7	13.1 [11.9, 14.3]	13.3	68.8 [65.5, 72.1]	69.4
	1.5	**50.3** [47.8, 52.6]	**50.7**	**18.0** [16.5, 19.3]	**18.3**	**14.8** [13.6, 16.0]	**15.0**	**71.9** [68.6, 75.2]	**72.5**
Audiobook (Trans)	**0.3**	44.5 [42.0, 46.8]	44.9	13.8 [12.3, 15.1]	14.1	10.7 [9.5, 11.9]	10.9	63.2 [59.9, 66.5]	63.8
	0.6	46.6 [44.1, 48.9]	47.0	15.1 [13.6, 16.4]	15.4	12.0 [10.8, 13.2]	12.2	65.9 [62.6, 69.2]	66.5
	1.0	48.8 [46.3, 51.1]	49.2	16.7 [15.2, 18.0]	17.0	13.4 [12.2, 14.6]	13.6	69.5 [66.2, 72.8]	70.1
	1.5	**51.0** [48.5, 53.3]	**51.4**	**18.3** [16.8, 19.6]	**18.6**	**15.1** [13.9, 16.3]	**15.3**	**72.6** [69.3, 75.9]	**73.2**
CHILDES (LSTM)	**0.3**	43.8 [41.3, 46.1]	44.2	13.5 [12.0, 14.8]	13.8	10.4 [9.2, 11.6]	10.6	62.5 [59.2, 65.8]	63.1
	0.6	45.9 [43.4, 48.2]	46.3	14.8 [13.3, 16.1]	15.1	11.7 [10.5, 12.9]	11.9	65.2 [61.9, 68.5]	65.8
	1.0	48.1 [45.6, 50.4]	48.5	16.4 [14.9, 17.7]	16.7	13.1 [11.9, 14.3]	13.3	68.8 [65.5, 72.1]	69.4
	1.5	**50.3** [47.8, 52.6]	**50.7**	**18.0** [16.5, 19.3]	**18.3**	**14.8** [13.6, 16.0]	**15.0**	**71.9** [68.6, 75.2]	**72.5**
CHILDES (Trans)	**0.3**	44.5 [42.0, 46.8]	44.9	13.8 [12.3, 15.1]	14.1	10.7 [9.5, 11.9]	10.9	63.2 [59.9, 66.5]	63.8
	0.6	46.6 [44.1, 48.9]	47.0	15.1 [13.6, 16.4]	15.4	12.0 [10.8, 13.2]	12.2	65.9 [62.6, 69.2]	66.5
	1.0	48.8 [46.3, 51.1]	49.2	16.7 [15.2, 18.0]	17.0	13.4 [12.2, 14.6]	13.6	69.5 [66.2, 72.8]	70.1
	1.5	**51.0** [48.5, 53.3]	**51.4**	**18.3** [16.8, 19.6]	**18.6**	**15.1** [13.9, 16.3]	**15.3**	**72.6** [69.3, 75.9]	**73.2**

TABLE 2.5 – Metric scores across different model settings presented in percentages (%). Mean values are averaged across different months, followed by range across different models. We also reported the score of the 36th month. TTR: Type-token ratio, NWType: Non-word type rate, NWToken: Non-word token rate. CHILDES child production. Best scores for each setting are in **bold**.

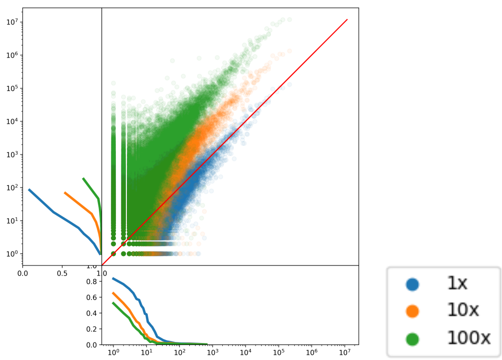


FIGURE 2.7 – **Word count distribution(temperature = 1.0)**.Word frequency relationships between training and generation sets under different sampling conditions (1x, 10x, 100x generation volume). x-axis: word frequency in the training set; y-axis: word frequency in the generation set. Side plot on the bottom show Missing rates as a function of word count in input corpus.

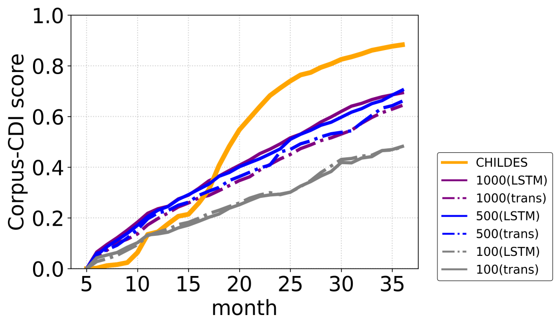


Figure 2.8 – **Model vocabulary growth across estimated annual exposure levels.** Comparison of corpus-CDI trajectories for models trained on 100-, 500-, and 1000-hour exposure estimates, alongside CHILDES child production data. Models show roughly linear vocabulary growth that scales modestly with total exposure but remains far below the human developmental curve.

Chapitre 3

Does episodic memory help close the lexical frequency gap in sensitivity to syntactic contrasts? A test using retrieval-augmented language models

Objectifs

Test whether retrieval-augmented language models (k-nearest-neighbor LMs) reduce the frequency gap that vanilla LMs show on syntactic minimal-pair contrasts, and identify which retrieval signals (structural vs. semantic) drive any gains.

3.1 Introduction

Language models are known to exhibit systematic lexical frequency biases: their performance on linguistic evaluations correlates with how often the lexical items in the test examples appear in training data, leading to low-frequency items posing a persistent challenge [Wei, 2021 ; Kim, 2022 ; Diehl Martinez, 2024]. While frequency does influence human performance in empirical tests of grammatical knowledge such as acceptability judgments [**lau2017grammaticality** ; Kempen, 2005 ; Bader, 2010, i.a.], recent work [Christensen, 2024] has shown evidence for robustness to lexical frequency in delineating the critical contrasts. What accounts for this gap between human robustness and model fragility?

One possible answer comes from Complementary Learning Systems (CLS) theory [McClelland, 1995 ; Kumaran, 2016], which proposes that the brain employs two distinct memory systems: a hippocampal system that rapidly encodes and retrieves specific experiences, and a neocortical system that gradually extracts generalizable statistical patterns. When parametric knowledge is weak, as it is for rare items, episodic memory can compensate by directly retrieving relevant past encounters [Davis, 2009]. This division of labor has also been proposed to be relevant to linguistic processing [Borensztajn, 2011 ; Duff, 2012], leading us to hypothesize that episodic retrieval may reflect the mechanism that underlies human robustness to lexical frequency in syntactic processing.

In this work, we computationally test this hypothesis by asking whether augmenting language models with an explicit retrieval mechanism helps close the lexical frequency gap in syntactic contrast evaluations. We adopt k-nearest-neighbor language models (kNN-LM; [Khandelwal, 2020]) as our proxy for episodic memory: the non-parametric datastore caches training instances for direct retrieval at inference time (analogous to hippocampal episodic traces), while the underlying parametric network consolidates statistical patterns through training (analogous to neocortical learning). The original kNN-LM has already been shown to be particularly helpful for rare patterns [Mallen, 2023 ; Kandpal, 2023], and recent work shows that the information available in the context can be used more flexibly than the same information stored parametrically [**chan2022transformers** ; Lampinen, 2025], both consistent with the CLS prediction. At the same time, a recent finding that kNN-LMs do *not* improve prediction of low-frequency tokens in standard language modeling evaluation [Nishida, 2025] makes it a non-trivial and empirically open question whether retrieval benefits extend to the domain of syntactic knowledge.

We investigate this question through syntactic contrast tests across three different phenomena (subject-verb agreement, wh-questions, relative clauses) using frequency-stratified test items, under both child-realistic and large-scale pretraining. We further analyze what drives retrieval effectiveness by contrasting full versus semantics-only retrieval, by examining the role of structural similarity in retrieved instances, and by varying the configurations of retrieval (retrieval granularity, context window size, and number of retrieved neighbors). Our results show that retrieval augmentation consistently narrows the lexical frequency gap across all phenomena and both model scales, with greater gains for more syntactically complex phenomena. Nevertheless, the gap remains not fully closed. To this end, we analyze the limitations of the current modeling approach and posit preferential reweighting, improved representations and retrieval for structural information, and flexible configurations of storage and retrieval as promising future directions.

3.2 Related Work

Frequency effects on linguistic evaluation in language models. Language models exhibit systematic frequency biases, where performance on linguistic evaluation often degrades when the test items contain low-frequency lexical items [Wei, 2021 ; Kim, 2022]. In the context of syntactic contrasts, models perform worse on minimal pairs that contain low-frequency words, even when the underlying grammatical contrast is identical to high-frequency cases. Diehl Martinez et al. [Diehl Martinez, 2024] directly quantify this frequency bias in BLiMP (one of our syntactic test datasets) and propose a training-time intervention (Syntactic Smoothing) to reduce it, finding that substantial gaps persist even after mitigation.

Wettig et al. [Wettig, 2023] examine how the masking rate during training determines which co-occurring token pairs (measured via pointwise mutual information) are exposed to the model, shaping both convergence speed and final performance. Since pointwise mutual information is itself derived from corpus-level token frequencies, these findings are consistent with frequency distributions in pretraining data influencing model behavior, though this connection is inferential. More directly, Kandpal et al. [Kandpal, 2023] show that language models struggle to acquire long-tail knowledge whose supporting documents are rare in the pretraining corpus, establishing a clear link between corpus frequency and what is learned.

Together, these findings establish that (lexical) frequency creates differential strength in parametric representations, with low-frequency items learned far less robustly. Our work asks a related question from a different methodological angle: rather than reducing frequency bias through better training, can it be compensated for at inference time through retrieval?

Complementary Learning Systems theory. Complementary Learning Systems (CLS) theory [McClelland, 1995 ; Kumaran, 2016] proposes that the human brain employs two specialized memory systems: a hippocampal system that rapidly encodes specific episodes, and a neocortical system that gradually extracts statistical regularities across experiences. This allows new information to be acquired quickly without disrupting slower cortical learning, and computational implementations of CLS show that it enables both rapid memorization and gradual generalization [McClelland, 1995]. Critically for our purposes, Eichenbaum [Eichenbaum, 2001] argues that the hippocampus does not merely store episodes but actively supports flexible use of past experiences through retrieval, enabling inferences that parametric learning alone may fail to achieve. Applied to language models, this predicts that when parametric representations are weak (as for low-frequency items), episodic retrieval should provide compensatory benefits by making specific relevant experiences directly accessible. Lampinen et al. [Lampinen, 2025] in particular demonstrates that language models fail to generalize certain types of knowledge learned parametrically (e.g., relation reversals) but succeed when the same information is available in context, suggesting that episodic retrieval enables more flexible use of past experiences than parametric consolidation. Similarly, studies of in-context learning show that Transformers can apply procedures to information presented in context that they cannot apply to parametrically-stored knowledge [**chan2022transformers**]. Our work extends these insights to syntactic processing by testing whether retrieval enables more robust generalization across frequency bands than parametric-only inferences.

Retrieval-augmented language models. Retrieval-augmented methods enhance language models by supplementing parametric knowledge with information drawn from an explicit memory at inference time. kNN-LMs [Khandelwal, 2020], which we adopt in this work, interpolate the base model's next-token distribution with a distribution computed from nearest-neighbor contexts retrieved from a datastore of training instances. While prior work shows benefits for rare patterns and knowledge-intensive tasks [Borgeaud, 2022; Kandpal, 2023; Mallen, 2023], recent work finds that standard kNN-LMs do not improve prediction performance for low-frequency tokens in language modeling evaluation [Nishida, 2025], leaving open questions about retrieval benefits for linguistic processing.

3.3 Method

We first show that frequency gaps in tests of grammatical knowledge exist in vanilla language models ("base models"), and then test whether kNN-LMs help close these gaps. To this end, we use targeted syntactic contrast datasets where we divide test instances based on the frequency of the content words they contain. We describe the details of the models, datasets, evaluation tasks and the retrieval procedure below.

3.3.1 Models and Training

Base models. We evaluate two decoder-only Transformer models that vary in parameter size and training data scale: (1) GPT-2 XL (1.5B parameters), pretrained on large-scale web text [Radford, 2019], and (2) GPT-2 Small (124M parameters) pretrained on child-directed speech corpora with 50M words selected from the BabyLM Challenge [Hu, 2024]. Both use a standard autoregressive objective and serve as parametric-only baselines. Further details are provided in Appendix 3.8.1.

***k*NN language models.** Following Khandelwal et al. [Khandelwal, 2020], we augment base models with a datastore $\mathcal{D}$ containing context–target pairs. For each token x_t, we store the key-value pair: $(\boldsymbol{f}_\theta(\boldsymbol{x}_{<t}), x_t)$, where $\boldsymbol{f}_\theta(\boldsymbol{x}_{<t})$ is the layer-normalized contextual representation fed as input to the feed-forward sublayer of the final Transformer block. For GPT-2, this corresponds to the output of the final block's second layer-normalization module and the input to its MLP. At inference time, we:

1. Compute the query: $\boldsymbol{q}_t = \boldsymbol{f}_\theta(\boldsymbol{x}_{<t})$
2. Retrieve k nearest neighbors via squared L2 distance
3. Compute the kNN probability: $p_{\mathrm{kNN}}(x_t|\boldsymbol{x}_{<t}) \propto \sum_{(\boldsymbol{k},v)\in\mathcal{N}} \mathbb{1}_{x_t=v} \exp(-d(\boldsymbol{k}, \boldsymbol{q}_t)/\tau)$
4. Interpolate: $p(x_t|\boldsymbol{x}_{<t}) = \lambda p_{\mathrm{kNN}} + (1-\lambda)p_{\mathrm{LM}}$

We fix $\lambda = 0.25$ following Khandelwal et al. [Khandelwal, 2020] and vary $k \in \{1, 16, 1024\}$, using FAISS [Johnson, 2019] for retrieval.

3.3.2 Evaluation Task

Syntactic contrasts. We evaluate sensitivity to three syntactic phenomena: Subject-Verb agreement (SV), Wh-questions (Wh), and Relative Clauses (RC). We collect examples in these categories from BLiMP [Warstadt, 2020], Zorro [Huebner, 2021a], and BIG-bench [Srivastava, 2023], filtering examples containing out-of-vocabulary items for each model. Each test item is a minimal pair with one grammatical and one ungrammatical sentence, where models must assign a higher probability to the grammatical sentence.

Frequency stratification. To test our hypothesis regarding lexical frequency, we further divided the evaluation data based on mean frequency across all the content words in the sentences (*high-frequency* ($f > 10^4$) versus *low-frequency* ($f < 10^3$); everything in-between *mid-frequency*). Our main hypothesis concerns the contrast between the high- and low-frequency bands; we report the mid-frequency band throughout to give a complete picture of how retrieval benefits scale with frequency.

We emphasize that all frequency gap claims are scoped *within* a single model/corpus: each gap and its narrowing under retrieval is measured against the same model's own baseline. Because the two pretraining corpora differ substantially in size and composition, the absolute frequency thresholds do not render items in the same band directly comparable across pretraining settings, and we therefore do not compare numerical trends across the two settings.

3.3.3 Retrieval Procedure

We construct three datastore variants to disentangle the contributions of structural and semantic information: (1) *Full Retrieval*: standard contextualized embeddings encoding both semantic and syntactic information; (2) *Semantics-only (averaged)*: position-averaged contextualized embeddings from (1);[1] and (3) *Semantics-only (uncontextualized)*: summed GloVe [Pennington, 2014] and fastText [Bojanowski, 2017] embeddings, capturing lexical semantics without structural information. We additionally vary three retrieval configuration parameters: granularity of retrieved instances (token, phrase, or sequence), context window size (τ), and number of retrieved neighbors (k), to characterize which retrieval configurations are most effective across syntactic phenomena.

Table 3.1 – Syntactic contrast accuracy and perplexity (PPL) for baseline and kNN-LM models ($k = 16$, $\tau = 3$, sequence-level retrieval). kNN augmentation improves accuracy across all phenomena and both pretraining scales, with relative clauses showing the largest gains. Perplexity reductions are larger for the large-scale model. Bold indicates improvement over baseline.

	Child-realistic		**Large-scale**	
	baseline	kNN	baseline	kNN
PPL. ↓	14.88	**14.30**	13.68	**11.60**
SV ↑	0.76	**0.82**	0.84	**0.89**
Wh ↑	0.77	**0.81**	0.81	**0.86**
RC ↑	0.68	**0.78**	0.65	**0.84**

1. In practice, averaging contextualized embeddings may not fully scrub syntactic information; we return to this issue later.

3.4 Results

3.4.1 Retrieval Benefits Syntactic Contrasts

We begin by investigating the general effectiveness of retrieval augmentation on syntactic contrasts. Table 3.1 [2] shows that kNN augmentation improves performance across all three syntactic phenomena and both pretraining scales, as well as replicating previously reported perplexity improvements [Khandelwal, 2020]. Relative clauses benefit the most from retrieval augmentation across both model scales. This pattern holds consistently regardless of pretraining scale, though perplexity reductions are larger for the large-scale model. The disproportionate benefit for relative clauses suggests that retrieval is most helpful where parametric representations are weakest, which motivates us to analyze the performance gains stratified by frequency.

3.4.2 Frequency Gap is Narrowed but not Fully Closed

Next, we examine whether retrieval helps close the lexical frequency gap. Table 3.2 confirms the existence of frequency gaps in both baseline models across three syntactic phenomena, with performance on low-frequency items consistently lower than on high-frequency items, and mid-frequency items falling in between. With retrieval augmentation, the high–low performance gap narrows across all phenomena and both model scales. In terms of error reduction ($\Delta\text{acc}/(1-\text{acc}_{\text{base}})$), low-frequency items show substantial relative gains (e.g., 49.2% for RCs in the large-scale model), though we note that this metric can be misleading near ceiling, where small absolute changes in already-high accuracy produce large negative percentages (e.g., child-realistic high-frequency Wh, 0.92 → 0.88, yields −50%). Raw accuracies should be used to adjust the interpretations in such cases.

TABLE 3.2 – Frequency-stratified syntactic contrast performance($k = 16$, $\tau = 3$; sequence-level retrieval). ER denotes error reduction, $\Delta\text{acc}/(1-\text{acc}_{\text{base}})$. Bold indicates improvement over baseline.

		Child-realistic			Large-scale		
		base	kNN	ER (%)	base	kNN	ER (%)
SV	high	0.91	0.90	−11.1	0.96	0.95	−25.0
	mid	0.82	**0.83**	5.6	0.90	0.89	−10.0
	low	0.60	**0.76**	40.0	0.71	**0.83**	41.4
Wh	high	0.92	0.88	−50.0	0.93	0.93	0.0
	mid	0.83	**0.86**	17.6	0.90	**0.91**	10.0
	low	0.61	**0.70**	23.1	0.69	**0.74**	16.1
RC	high	0.88	**0.89**	8.3	0.89	**0.95**	54.5
	mid	0.73	**0.81**	29.6	0.82	**0.89**	38.9
	low	0.48	**0.65**	32.7	0.41	**0.70**	49.2

Statistical analysis supports the interpretation that narrowing is driven primarily by gains on low-frequency items. Bootstrapped 95% confidence intervals on the per-frequency band performance delta (kNN − baseline) confirm a robust separation: for the child-realistic model, low-frequency items show $\Delta = 0.140$ (95% CI $[0.115, 0.166]$) versus $\Delta = -0.013$ (95% CI

2. Note for all tables in this section: Baseline models use parametric knowledge only; kNN models are retrieval-augmented. Abbreviations: SV (subject-verb agreement), Wh (wh-questions), RC (relative clauses), PPL (perplexity). We use $k = 16$, $\tau = 3$, and sequence-level retrieval (see Section 3.3.3) as the representative configuration, since this setting yields general gains in low-frequency conditions. However, as discussed in Section 3.5.2, there may exist alternative configurations that outperform this representative configuration for individual phenomena.

$[-0.030, 0.004]$) for high-frequency items, with non-overlapping intervals (large-scale model shows the same pattern; see Appendix 3.8.3).

A mixed-effects logistic regression using treatment coding with high-frequency items and the baseline model as reference levels (`correct ∼ Frequency * ModelType + (1|item) + (1|phenomenon)`) further shows that the retrieval effect is significantly larger for low-frequency than for high-frequency items (Frequency [low] × ModelType [kNN] interaction: $\beta = 0.52$, $p < 0.001$. The effect of retrieval within the high-frequency band is not significant (ModelType [kNN]: $\beta = -0.04$, $p = 0.62$; Appendix 3.8.3).

Nevertheless, this compensation is not sufficient to eliminate the frequency gap: low-frequency items still yield lower accuracy than high-frequency items across all tested phenomena and both model scales, even after retrieval augmentation.

3.5 Analysis

The analyses in this section fix the retrieval configuration to $k = 16$, $\tau = 3$ with sequence-level retrieval, which isolates the contribution of the factors studied from the effects of configuration.

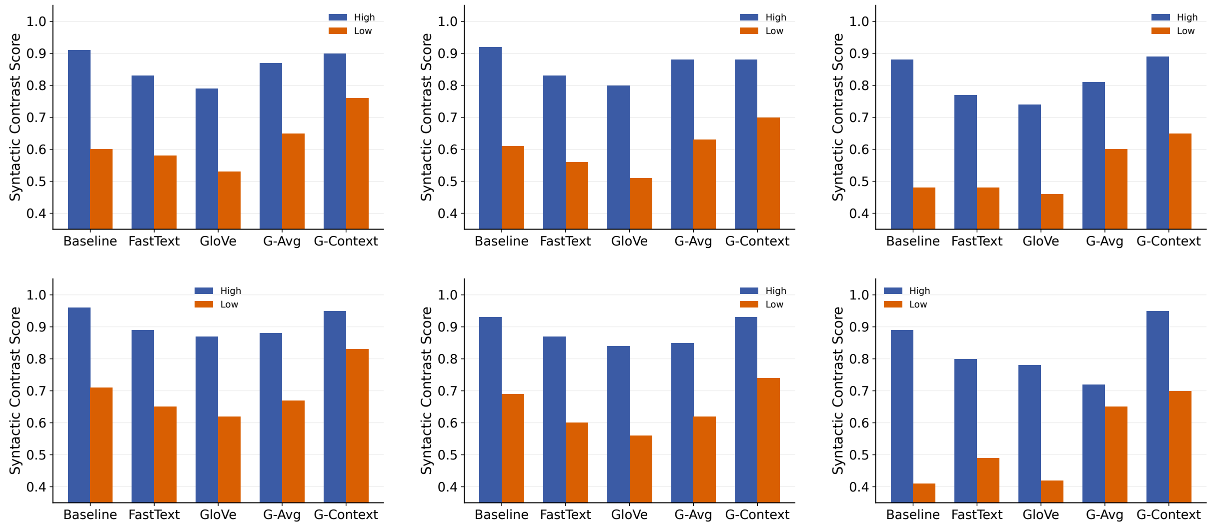
3.5.1 Importance of Structural vs. Semantic Information

We proceed to investigate which aspects of the retrieved information drive the observed benefits by examining (1) the effect of semantics-only retrieval, and (2) the effect of retrieving structurally similar instances.

Semantics-only retrieval is mostly ineffective. Figure 3.1 shows that semantics-only retrieval (FastText, GloVe, G-Avg; see Section 3.3.3) performs at or substantially *below* the baseline without retrieval across most conditions. This shows that semantic similarity of the retrieved examples alone does not reliably improve syntactic judgments. One partial exception is GPT2-Averaged (G-Avg), where retrieval does lead to substantive gains over the baseline for low-frequency items for RCs, although performance on high-frequency items degrades by a large margin. FastText and GloVe also yield gains for low-frequency RCs in the large-scale setting, although relatively minor. There are two possible, non-mutually exclusive interpretations for this: (1) semantic information does sometimes yield benefits, as discussed in the next paragraph; or (2) that averaging of contextualized embeddings does not totally remove structural information.

Structural information is critical, and similarity-based example selection does not compensate for missing structure. Figure 3.2 examines retrieval benefits from two angles: whether the set of retrieved items contains structurally matching examples (*with (w)* vs. *without (w/o)*),[3] and whether retrieval ranks them by embedding similarity or at random (*+r*). Across all phenomena and both model scales, the *w* bars consistently exceed *w/o* for low-frequency items (orange bars) with the strongest effects for relative clauses. This illustrates the importance of structure-matched examples in the retrieved set. For low-frequency items, both structural matches and similarity-based selection contribute. However, similarity does not compensate for missing

3. See Appendix 3.8.4 for our structural match metric.



(a) Subject-verb agreement (b) Wh-questions (c) Relative Clauses

Figure 3.1 – Performance on syntactic contrasts using semantics-only retrieval strategies compared to full retrieval ($k = 16$, $\tau = 3$) of the child-realistic (top row) and large-scale (bottom row) models. Blue bars indicate high-frequency items and orange bars low-frequency items.

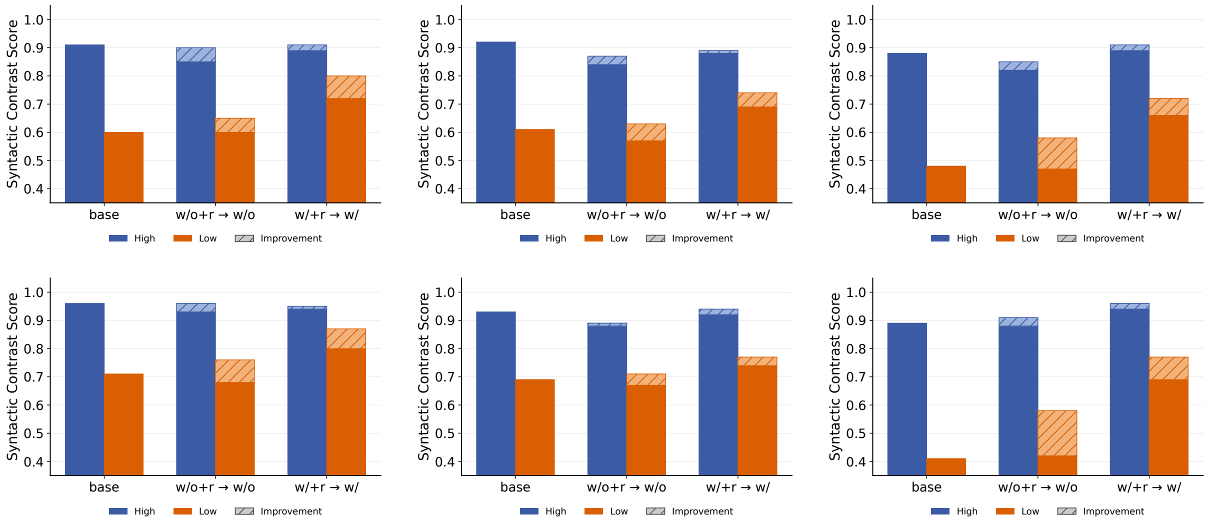
structural information. While it does add smaller, consistent gains overall, similarity-based selection without structural matches (*w/o*: middle orange bars with hatched gains) consistently fails to outperform random selection with structural matches (*w/+r*: right orange bars without hatched gains) for any model or phenomenon. Thus, structurally matched examples are the primary driver of frequency-selective retrieval benefits, with secondary contributions from similarity-based selection. Given the ineffectiveness of semantics-only retrieval discussed above, the embedding similarity gains likely reflect partial structural similarity between the retrieved examples and the target, in addition to semantic similarity.

3.5.2 Effects of Retrieval Configuration

We next analyze whether and how retrieval effectiveness is modulated by three configuration parameters: retrieval granularity (token, phrase, or sequence), number of neighbors (k), and context window size (τ).

Effects of retrieval granularity. Figure 3.3 shows that, at $k = 16$ and $\tau = 3$, sequence-level retrieval numerically provides the strongest or tied-strongest performance for low-frequency items across all phenomena and both model settings, but the effect of retrieval granularity overall is small.

Effects of context window size. Figure 3.4 shows that the effect of context window size is idiosyncratic. For RC, $\tau = 10$ yields the highest accuracy across both model scales and both frequency bands, whereas for SV, $\tau = 3$ yields the strongest performance for low-frequency. For Wh, the effect of τ is minimal, except $\tau = 3$ leading to degradation in high-frequency items in the child-realistic model. Thus, increasing context window size does not uniformly improve performance, and the preferred window size depends on the phenomenon, frequency band, and



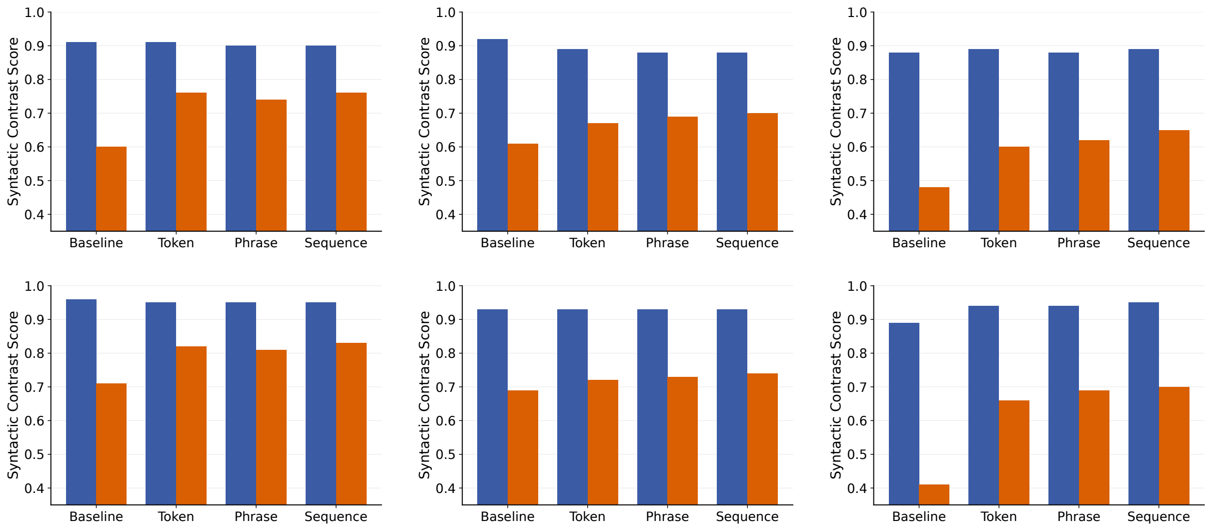
(a) Subject-verb agreement (b) Wh-questions (c) Relative clauses

FIGURE 3.2 – Performance when retrieved sequences contain target syntactic structures (with: w/) versus when they do not (without: w/o). Top row: child-realistic model; bottom row: large-scale model ($k = 16$, $\tau = 3$, sequence-level). Blue and orange bars correspond to high- and low-frequency items, respectively. Within each panel, the x-axis shows three conditions: **base** (no retrieval), **w/o+r** $\rightarrow$ **w/o** (retrieved set does not contain structurally matched examples, random selection $\rightarrow$ embedding-similarity), and **w/+r** $\rightarrow$ **w/** (retrieved set contains structurally matched examples, random selection $\rightarrow$ embedding-similarity). Solid bars denote random selection (**+r**); hatched segments denote the additional gain from embedding-similarity ranking over random selection.

model.

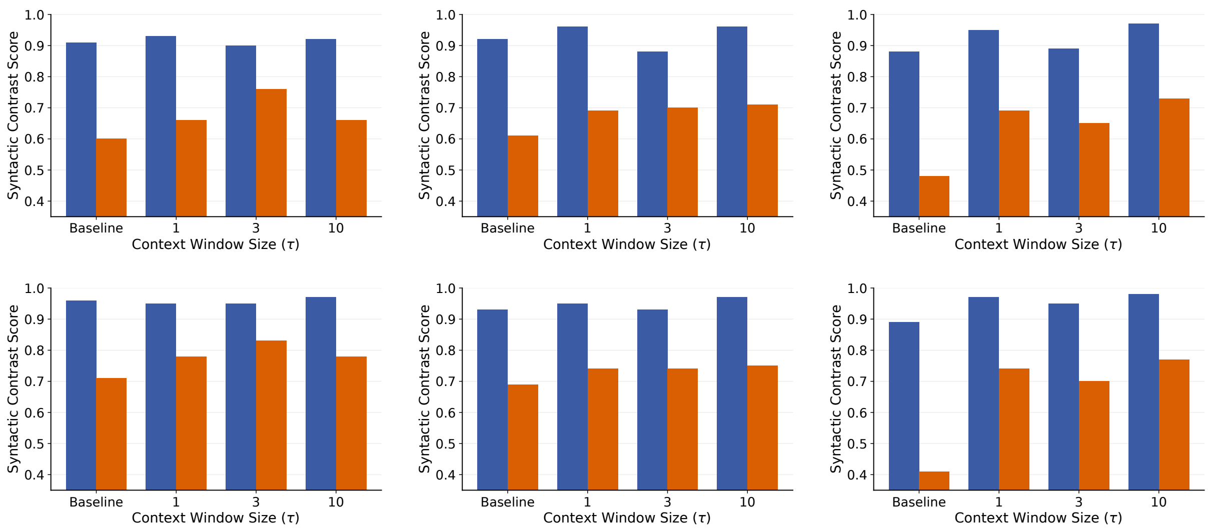
Effects of number of retrieved instances. Figure 3.5 shows that the number of retrieved instances has a non-monotonic effect on performance in the low-frequency regime. Moving from $k = 1$ to a moderate number ($k = 16$) consistently improves performance across most conditions, indicating the benefit of aggregating evidence from multiple exemplars. However, further increasing the number of neighbors ($k = 1024$) leads to performance degradation, likely due to noise from distant and less relevant neighbors. This pattern holds across models, granularities, and syntactic phenomena, suggesting the optimal choice lies at intermediate values of k. The effect for high-frequency items is smaller and mixed.

No cross-phenomena optimal configuration. Across the three parameters (retrieval granularity, context window size, and number of retrieved instances), no single configuration consistently yields optimal performance across all syntactic phenomena, model, and frequency strata. While sequence-level retrieval, larger context windows, and a moderate number of neighbors ($k = 16$) provide the most robust overall performance (and therefore serves as our representative configuration), it does not uniformly dominate across all settings, particularly for simpler or high-frequency dependencies where more localized representations or smaller contexts can be competitive. These results suggest that effective retrieval is jointly modulated by dependency complexity, data frequency, and model capacity, and that optimal retrieval configurations are likely to be phenomenon-dependent rather than universally fixed.



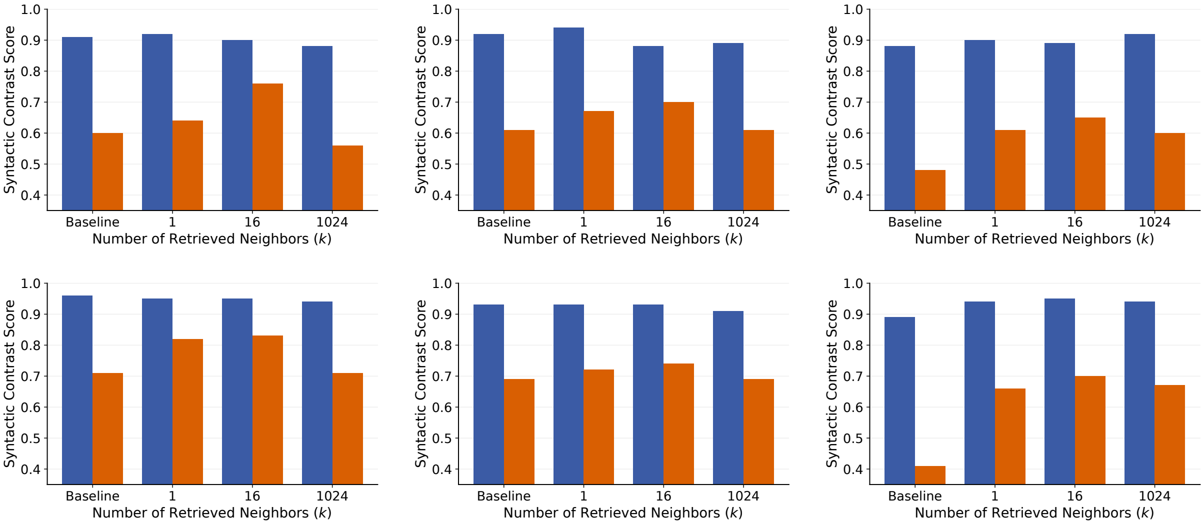
(a) Subject-verb agreement (b) Wh-questions (c) Relative clauses

FIGURE 3.3 – Performance across retrieval granularities (token, phrase, sequence) in child-realistic (top row) and large-scale (bottom row) models ($k = 16$, $\tau = 3$). Blue bars indicate high-frequency items and orange bars low-frequency items. Sequence-level retrieval is numerically the overall strongest, but the effect of retrieval granularity is overall small.



(a) Subject-verb agreement (b) Wh-questions (c) Relative clauses

FIGURE 3.4 – Performance across context-window sizes for the child-realistic (top row) and large-scale (bottom row) settings, using sequence-level retrieval with $k = 16$. Blue bars indicate high-frequency items and orange bars indicate low-frequency items. The effects of context size are non-monotonic and phenomenon-, frequency band- and model-dependent.



(a) Subject–verb agreement (b) Wh-questions (c) Relative clauses

FIGURE 3.5 – Performance across numbers of retrieved neighbors ($k \in \{1, 16, 1024\}$) for the child-realistic (top row) and large-scale (bottom row) settings, using $\tau = 3$ and sequence-level retrieval. Blue bars indicate high-frequency items and orange bars indicate low-frequency items. For all six low-frequency conditions, $k = 16$ outperforms both $k = 1$ and $k = 1024$. Effects on high-frequency items are smaller and mixed.

3.6 Discussion

We tested whether an episodic retrieval mechanism as implemented via kNN-LMs can help close the lexical frequency gap that language models exhibit in syntactic contrast tests. Our findings reveal that retrieval augmentation provides substantial compensatory benefits for low-frequency items, and that both structural and semantic information contribute to the gains, although the role of structural information is more critical. However, the compensation is insufficient: the frequency gap is narrowed but not fully closed. In this section, we discuss which modeling considerations may help narrow this gap further in the future based on the limitations we observe in our analyses. Specifically, our analyses suggest that preferential reweighting of retrieved instances, improved representations and retrieval strategies for structural information, and an adaptive storage and retrieval mechanism that allow for flexible configurations of instances may be promising next steps.

Preferential reweighting. The hippocampus has been proposed to engage in preferential reweighting, where infrequent (but significant) instances are prioritized in storage as well as retrieval [Kumaran, 2016]. This kind of modulation is consistent with error-driven learning accounts [Chang, 2006 ; Rescorla, 1972], which predict that unexpected or surprising items receive stronger consolidation. Standard kNN-LM as implemented in this study retrieves instances by similarity alone, without modulating retrieval by the rarity or informativeness of stored instances. This leads to suboptimal retrieval of relevant examples especially for more difficult phenomena like RCs in the child-realistic regime (e.g., the low rate of % structurally similar instances retrieved for RCs: Table **??**), indicating that improving strategies for retrieval and storage to identify better retrieval targets could lead to closing the frequency gap further. Recent computational work similarly finds benefits from selective data weighting [Toneva, 2019] and importance-based sampling [Xie, 2023], suggesting that incorporating reweighting strategies into the retrieval mechanism could amplify benefits in the low-frequency regime.

Better representations and retrieval strategies for structural information. Our analysis shows that structural information is critical for retrieval benefits we observe, which is in line with episodic accounts of syntactic processing that focus on the structural aspects of the retrieved content [Borensztajn, 2011]. In light of this observation, we note two potential issues that may contribute to the persistent frequency gap. First, as discussed above, retrieval based on representation similarity is not always effective in identifying structurally relevant instances, particularly in child-realistic models. Second, even when relevant instances are retrieved, the structural signals encoded in the representations may be too weak to fully boostrap processing of low-frequency instances. Addressing the first issue may benefit from selective indexing strategies and constraint-based filtering strategies [Borgeaud, 2022 ; Izacard, 2022], and the second may require expanding the datastore to maintain the full residual stream representations or exploring alternative pooling strategies in the datastore.

Flexible configurations of storage and retrieval. The absence of a universal retrieval configuration across different syntactic phenomena suggests that maximally effective episodic retrieval requires adaptive mechanisms that modulate granularity, context scope, and neighbor count. Indeed, human episodic memory has been proposed to support this kind of flexibility [Duff, 2012] and adaptive retrieval mechanisms grounded in metacognitive monitoring have been discussed in memory theory [Nelson, 1990]. Current kNN-LM uses a fixed configuration across all inputs. Developing retrieval strategies that adapt their configuration to the structural demands of the input represents a promising direction for closing the remaining frequency gap.

3.7 Conclusion

We tested whether an episodic retrieval mechanism can help close the lexical frequency gap that language models exhibit in syntactic contrast evaluations. Using kNN-LMs as a computational proxy for hippocampal episodic memory, we find consistent compensatory benefits for low-frequency items across syntactic phenomena and model/data scales, with structural information playing a critical role in driving these benefits. However, the frequency gap remains not fully closed—test items with low frequency lexical items still systematically lead to lower performance even after retrieval augmentation. The limitations of the current implementation of episodic memory identified through our analyses lead to three concrete future directions for improvement: implementation of a preferential reweighting mechanism, stronger structural representations and retrieval strategies, and adaptive retrieval configurations rather than fixed. More broadly, these results offer proof-of-concept support for the CLS hypothesis that episodic memory contributes to the robustness of syntactic processing to lexical frequency effects, while identifying areas for improvement for future modeling work.

3.8 Appendix

3.8.1 Pretrained models and datasets

Child-realistic pretraining data. We constructed a dataset to approximate the language input a child receives during early development, drawing on child-directed speech (CDS) from CHILDES [MacWhinney, 2000], transcribed conversational speech from the Switchboard Dialog Act Corpus [**stolcke2000dialogue** ; Godfrey, 1992], children's stories from the Children's Stories Text Corpus [**bensaid2021cstc** ; **bensaid2021fairytailor**], and movie and television subtitles from OpenSubtitles [**lison2016opensubtitles**]. These four sources also appear in the 2023 BabyLM corpus release [Warstadt, 2023]. The complete dataset comprises 50.03M words, consistent with estimates of cumulative linguistic input by early childhood [Gilkerson, 2017 ; Frank, 2023]. Table 3.3 shows the distribution of constituent sub-datasets.

Large-scale pretraining data. For the large-scale condition, we use GPT-2 XL (1.5B parameters) pretrained on the WebText corpus [Radford, 2019], which contains approximately 40GB of text scraped from internet sources.

Model architectures and training. Table 3.4 summarizes the architectural specifications for both models. The child-realistic model uses the GPT-2 Small architecture with 124M parameters, trained from scratch for 10 epochs, following the BabyLM Challenge setup [Hu, 2024]. The GPT-2

TABLE 3.3 – Composition of child-realistic pretraining corpus combining child-directed speech, conversational dialogue, and children's books to approximate naturalistic language exposure during early childhood. CDS = child-directed speech; SWBD = Switchboard; CSTC = Children's Stories Text Corpus; OSub = OpenSubtitles.

Dataset	**Domain**	**# words**	**Prop. (%)**
CHILDES	CDS	15M	30.0
SWBD	Dialogue	1.18M	2.4
CSTC	Books	3.22M	6.4
OSub	Dialogue	30.63M	61.2
Total		**50.03M**	**100.0**

XL model uses the publicly available pretrained checkpoint from RADFORD et al. [Radford, 2019]. The training details for the child-realistic model are provided in Table 3.5.

TABLE 3.4 – Model architecture specifications. GPT-2 Small was trained from scratch on child-realistic data (50M words); GPT-2 XL uses the publicly released pretrained checkpoint from RADFORD et al. [Radford, 2019] trained on WebText (∼40GB).

Parameter	**GPT-2 Small**	**GPT-2 XL**
# parameters	124M	1.5B
Transformer layers	12	48
Hidden size	768	1600
FFN size	3072	6400
Attention heads	12	25
Attention dropout	0.1	0.1
Embedding dimension	768	1600
Vocabulary size	50257	50257

3.8.2 Evaluation set construction

To ensure focused evaluation on syntactic phenomena relevant to our research questions, we selected subsets corresponding to our target phenomena from three evaluation benchmarks: BLiMP [Warstadt, 2020], Zorro [Huebner, 2021a], and BIG-bench [Srivastava, 2023] as discussed below.

3.8.2.1 Source benchmarks

BLiMP. The Benchmark of Linguistic Minimal Pairs for English [Warstadt, 2020] consists of 67 sub-datasets, each containing 1,000 minimal pairs (67,000 pairs total) isolating specific contrasts in syntax, morphology, or semantics. The data is automatically generated according to linguist-crafted grammar templates, with 96.4% aggregate human agreement with the labels. For subject–verb agreement, we selected the relational-noun distractor paradigm and four plural-agreement paradigms: regular and irregular plurals, each with a verb-number manipulation

TABLE 3.5 – Training hyperparameters for GPT-2 Small on the child-realistic corpus, trained for 10 epochs using a standard autoregressive language modeling objective with cross-entropy loss.

Hyperparameter	**Value**
Max sequence length	1024
Batch size	64
Training epochs	10
Learning rate	0.0001
LR scheduler	inverse sqrt
Warmup steps	1000
Optimizer	Adam
Adam-β_1	0.9
Adam-β_2	0.98
Dropout	0.1
Weight decay	0.01
Gradient clipping	1.0

holding the subject fixed (variant 1) and a subject-number manipulation holding the verb fixed (variant 2). For wh-questions, we selected the object-gap, subject-gap, and long-distance subject-gap wh-question paradigms, together with the four wh-versus-that paradigms crossing gap presence and dependency length. For relative clauses, we selected the relative-clause distractor agreement paradigm.

Zorro. Zorro [Huebner, 2021a] is a targeted grammar test suite designed for small-data, restricted-vocabulary evaluation. It comprises 23 paradigms spanning 13 grammatical phenomena, with 2,000 minimal pairs (4,000 sentences) per paradigm. Its lexical items were selected from whole-word entries in BabyBERTa's custom vocabulary, making the suite suitable for evaluating models trained on comparatively limited, child-directed linguistic input. For subject–verb agreement, we selected agreement across a prepositional phrase, agreement in questions with an auxiliary, and agreement in simple questions. For wh-questions, we selected the object and subject wh-question paradigms. For relative clauses, we selected agreement across a relative clause.

BIG-bench. The Beyond the Imitation Game Benchmark [Srivastava, 2023] includes several linguistic evaluation tasks. For subject-verb agreement, we selected the simple-English, single-adverb, double-adverb, adverb-conjunction, name-plus-prepositional-phrase, noun-plus-prepositional-phrase, noun-plus-prepositional-phrase-and-adverb, Gulordava-nonce, and frequency-disbalance-nonce subtasks. For relative clauses, we selected the long-nested-inner and long-nested-outer subtasks. No BIG-bench items were used for the wh-question evaluation.

3.8.3 Statistical analyses

Bootstrapped confidence intervals on per-frequency band deltas. For each frequency band, we computed the performance delta (kNN − baseline) and estimated 95% confidence intervals via bootstrap resampling over test items (10,000 resamples). As seen in Table 3.6, in both settings, the intervals for low- and high-frequency items do not overlap, indicating a robust separation in retrieval gains.

Table 3.6 – Bootstrapped 95% confidence intervals on the per-frequency band accuracy delta (kNN − baseline), 10,000 resamples over test items ($k = 16$, $\tau = 3$, sequence-level).

Pretraining	Freq	Δ	95% CI
	high	−0.013	[−0.030, 0.004]
Child-realistic	mid	*0.040*	[0.022, 0.059]
	low	*0.140*	[0.115, 0.166]
	high	*0.017*	[0.002, 0.033]
Large-scale	mid	*0.023*	[0.008, 0.039]
	low	*0.153*	[0.125, 0.182]

Mixed-effects logistic regression. To test whether the effect of retrieval is larger for low-frequency items, we fit a mixed-effects logistic regression predicting correctness from Frequency (high/low) and Model Type (baseline/kNN), with random intercepts for item and for phenomenon. We used treatment coding, with high-frequency items and the baseline model as the reference levels:

$$\texttt{correct} \sim \texttt{Frequency * ModelType + (1|item) + (1|phenomenon)}$$

Coefficients are reported in Table 3.7. Under this coding, the Model Type coefficient estimates the effect of kNN augmentation within the high-frequency band. This coefficient was not statistically significant ($\beta = -0.04$, $p = 0.62$), indicating no detectable retrieval effect for high-frequency items. Crucially, the Frequency × Model Type interaction was positive and significant ($\beta = 0.52$, $p < 0.001$), indicating that the retrieval effect (in log odds) was larger for low-frequency than for high-frequency items. Together with the positive accuracy differences and bootstrap confidence intervals for low-frequency items in Table 3.6, this result supports the hypothesis that retrieval disproportionately benefits items for which parametric representations are weak, rather than providing a uniform improvement across frequency bands.

Table 3.7 – Mixed-effects logistic regression coefficients (`correct` $\sim$ `Frequency * ModelType + (1|item) + (1|phenomenon)`; $k = 16$, $\tau = 3$, sequence-level).

Term	β	SE	z	p
Frequency (low)	−1.52	*0.11*	−13.8	< 0.001
ModelType (kNN)	−0.04	*0.08*	−0.50	*0.62*
Frequency × ModelType	*0.52*	*0.10*	*5.20*	< 0.001

3.8.4 Structural match analysis

For each evaluation item and retrieved sequence, we defined a binary structural match indicator equal to 1 if the sequence contained the same target phenomenon—subject-verb agreement (SV), a wh-question (Wh), or a relative clause (RC)—and 0 otherwise. We identified these phenomena using spaCy [Honnibal, 2020] as specified below:

Subject-Verb detection. We applied spaCy's dependency parser to identify subject-verb relationships. Instances were flagged if they contained `nsubj` (nominal subject) or `nsubjpass` (passive nominal subject) dependencies that link a noun phrase to a finite verb.

Wh-Sentence detection. We used part-of-speech (POS) tagging to identify wh-words (POS tags: `WDT`, `WP`, `WP$`, `WRB`) at clause boundaries. Instances containing interrogative or relative wh-elements were marked for inclusion.

Relative Clause detection. We combined POS tagging with dependency parsing to identify relative clause structures. Specifically, we detected: (1) Relative pronouns (`WDT`, `WP`) or relativizers (*that*, *which*, *who*); (2) Clausal complements with `relcl` (relative clause modifier) dependency relations; and (3) Appropriate structural embedding within a noun phrase.

Table 3.8 reports the mean proportion of the 16 retrieved neighbors that contain the target syntactic structure, and Table 3.9 reports two-sample t-tests comparing the effect of retrieval with vs. without structural matches, each against the baseline and against each other.

TABLE 3.8 – Mean proportion of the 16 retrieved neighbors containing the target syntactic structure ($k = 16$, $\tau = 3$, sequence-level retrieval). Structural matches remain particularly sparse for relative clauses in the child-realistic setting.

	SV		**Wh**		**RC**	
	high	low	high	low	high	low
Child-realistic	0.74	0.70	0.49	0.62	0.09	0.15
Large-scale	0.95	0.89	0.67	0.76	0.74	0.70

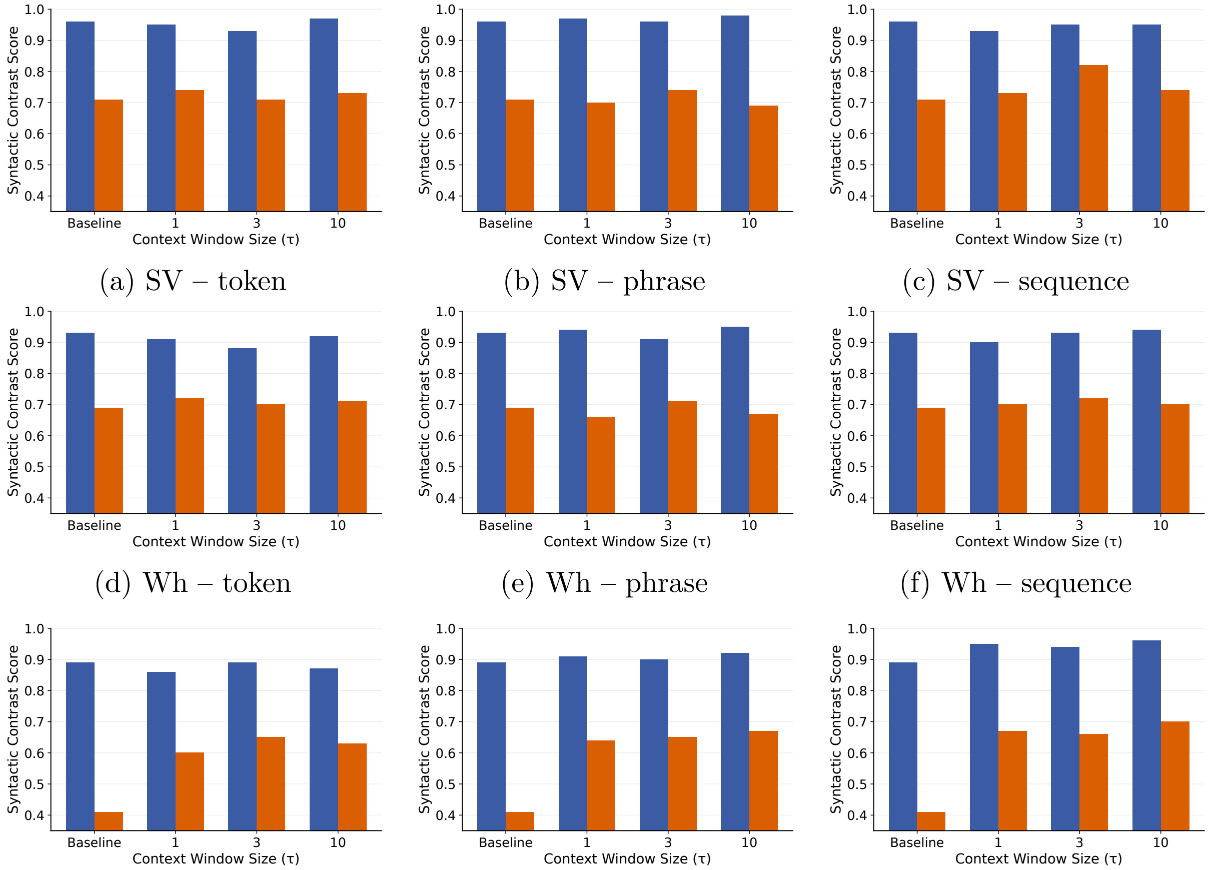
TABLE 3.9 – p-values for structural-retrieval comparisons in the child-realistic and large-scale models ($k = 16$, $\tau = 3$, sequence-level retrieval).

		Child-realistic			**Large-scale**		
Comparison	Freq	SV	Wh	RC	SV	Wh	RC
Baseline vs. Without	low	0.16	0.38	0.022	0.071	0.49	< 0.001
	high	0.67	0.14	0.43	0.76	0.008	0.28
Baseline vs. With	low	0.006	0.021	< 0.001	< 0.001	0.004	< 0.001
	high	0.36	0.23	0.016	0.44	0.52	< 0.001
With vs. Without	low	0.003	0.046	< 0.001	< 0.001	0.031	< 0.001
	high	0.31	0.078	0.012	0.44	0.001	0.001

The tests support the claim that retrieval benefits are greater when the retrieved examples contain structurally matched examples, and reveal additional nuance: the structural-match advantage is more pronounced in low-frequency bands and for structurally complex phenomena (RCs), consistent with broader trends elsewhere in the manuscript.

3.8.5 Additional Retrieval Configuration Results

This section provides additional results for the retrieval configuration analyses in the main text. All results use the *Full Retrieval* datastore defined in Section 3.3.3; baseline results are shown for reference. Figures 3.6–3.9 vary context-window size ($\tau \in \{1, 3, 10\}$) at $k \in \{1, 1024\}$, whereas Figures 3.10–3.13 vary the number of retrieved neighbors ($k \in \{1, 16, 1024\}$) at $\tau \in \{1, 10\}$. Together with the main analyses, these figures cover the evaluated combinations of neighbor count, context-window size, and retrieval granularity for the displayed high- and low-frequency



(a) SV – token (b) SV – phrase (c) SV – sequence

(d) Wh – token (e) Wh – phrase (f) Wh – sequence

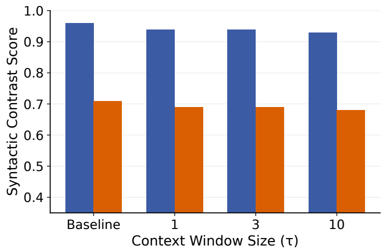
(g) RC – token (h) RC – phrase (i) RC – sequence

FIGURE 3.6 – Effect of context window size ($\tau \in \{1, 3, 10\}$) for GPT-2 XL at $k = 1$. Blue bars indicate high-frequency items and orange bars indicate low-frequency items; baseline results are shown for reference.

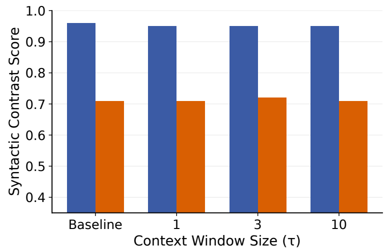
bands in both pretraining settings. Given many differences between individual configurations are small, we focus here on broad patterns rather than numerical rankings within individual panels.

Effects of context window size. Figures 3.6–3.9 extend Figure 3.4 to $k \in \{1, 1024\}$ and to all retrieval granularities. As in the main analysis, accuracy is non-monotonic with context window size. The direction and magnitude of the effects vary by number of neighbors, retrieval granularity, phenomenon, and frequency band; and are often small. These additional results therefore do not identify a context window size that is uniformly preferred across conditions.

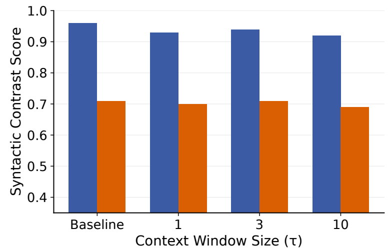
Effects of the number of retrieved neighbors. Figures 3.10–3.13 extend Figure 3.5 to $\tau \in \{1, 10\}$ and to all retrieval granularities. The results are broadly consistent with the pattern reported in the main text: for low-frequency items, the intermediate setting $k = 16$ generally yields the strongest performance among the tested values, whereas effects on high-frequency items are smaller and more mixed. The magnitude and direction of the effects vary across conditions, and the effects are often small. We therefore treat $k = 16$ as a useful representative setting rather than as a universally optimal neighbor count. Comparisons between $\tau = 1$ and $\tau = 10$ similarly provide no evidence for an interesting interaction effect with context window size.



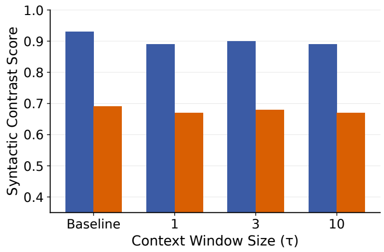
(a) SV – token



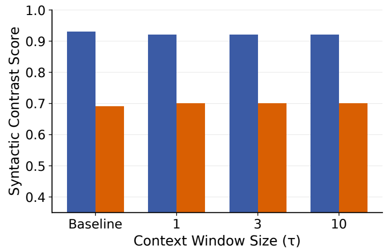
(b) SV – phrase



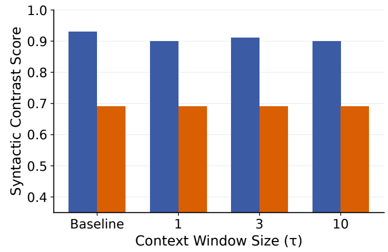
(c) SV – sequence



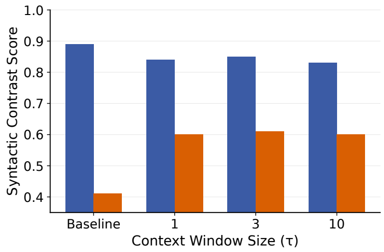
(d) Wh – token



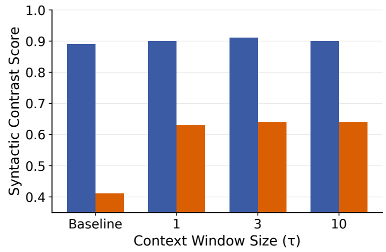
(e) Wh – phrase



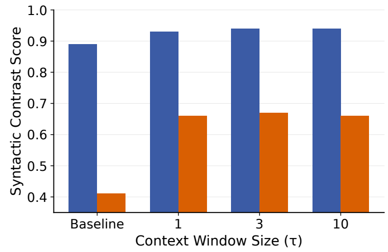
(f) Wh – sequence



(g) RC – token

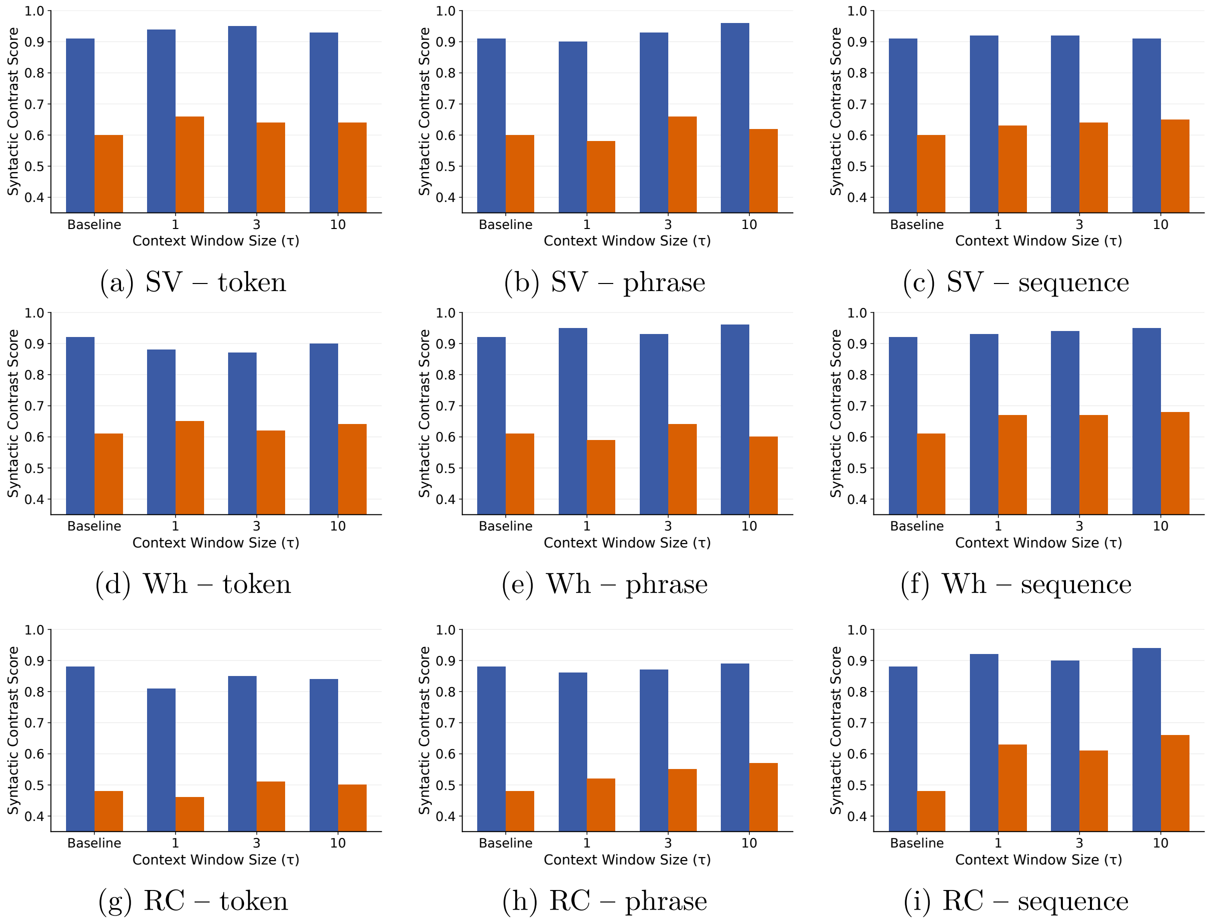


(h) RC – phrase



(i) RC – sequence

FIGURE 3.7 – Effect of context window size ($\tau \in \{1, 3, 10\}$) for GPT-2 XL at $k = 1024$. Blue bars indicate high-frequency items and orange bars indicate low-frequency items; baseline results are shown for reference.

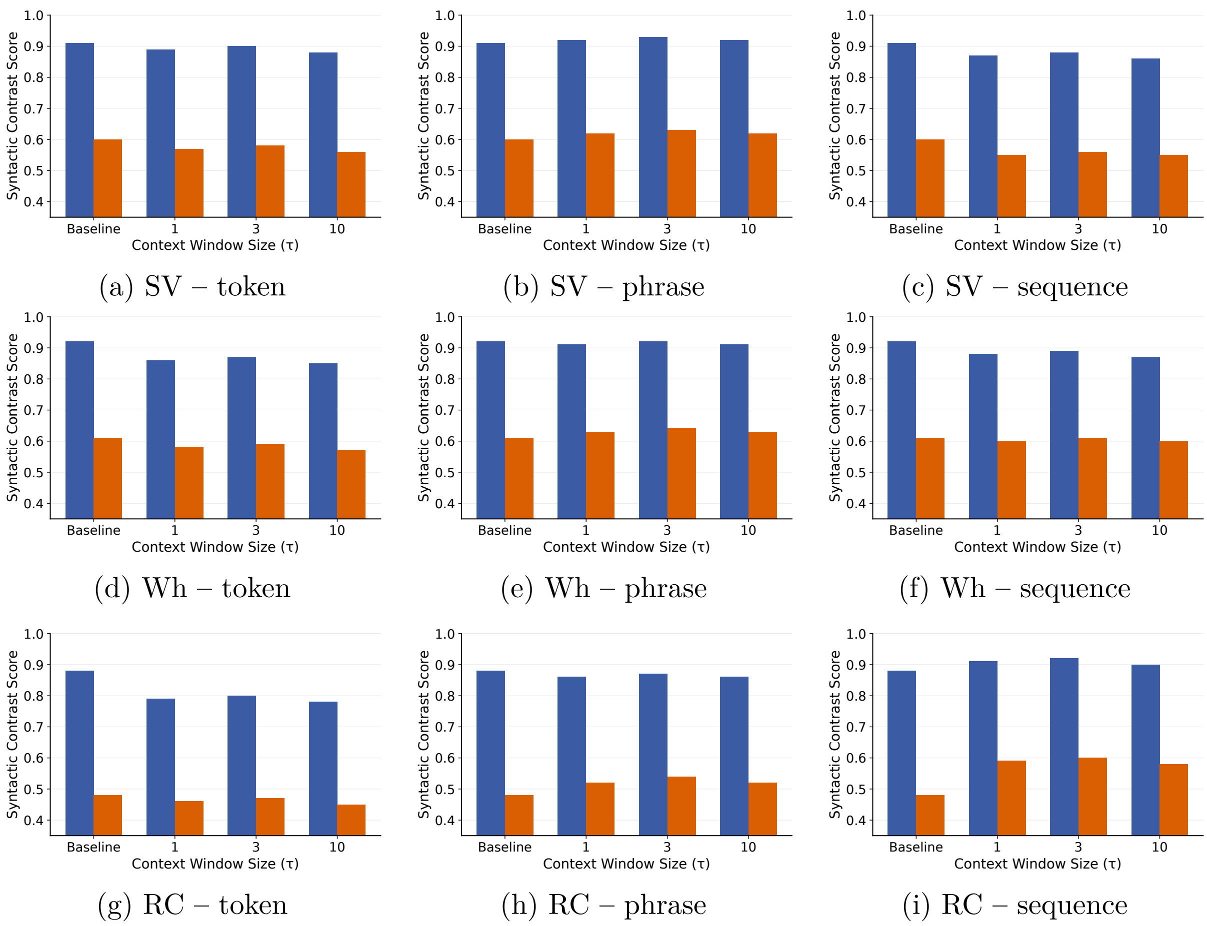


(a) SV – token (b) SV – phrase (c) SV – sequence

(d) Wh – token (e) Wh – phrase (f) Wh – sequence

(g) RC – token (h) RC – phrase (i) RC – sequence

FIGURE 3.8 – Effect of context window size ($\tau \in \{1, 3, 10\}$) for the child-realistic model at $k = 1$. Blue bars indicate high-frequency items and orange bars indicate low-frequency items; baseline results are shown for reference.

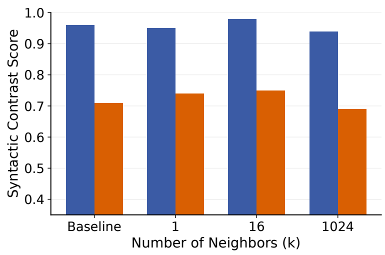


(a) SV – token (b) SV – phrase (c) SV – sequence

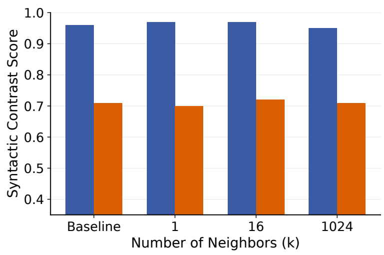
(d) Wh – token (e) Wh – phrase (f) Wh – sequence

(g) RC – token (h) RC – phrase (i) RC – sequence

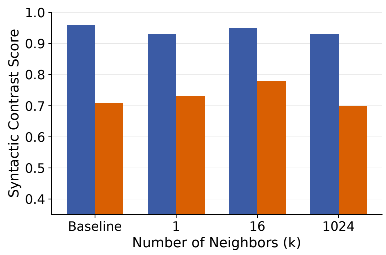
FIGURE 3.9 – Effect of context window size ($\tau \in \{1, 3, 10\}$) for the child-realistic model at $k = 1024$. Blue bars indicate high-frequency items and orange bars indicate low-frequency items; baseline results are shown for reference.



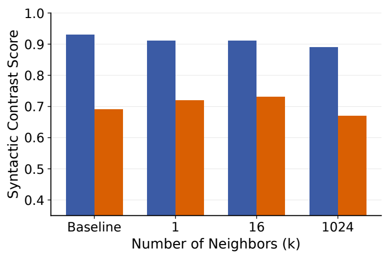
(a) SV – token



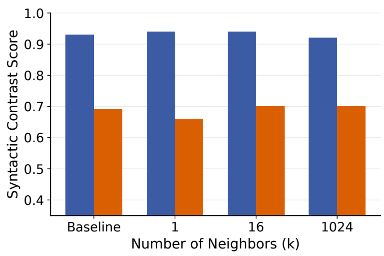
(b) SV – phrase



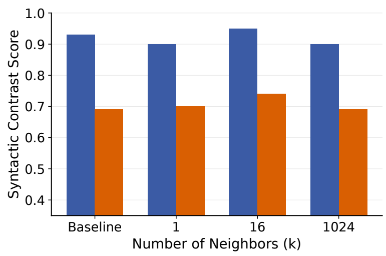
(c) SV – sequence



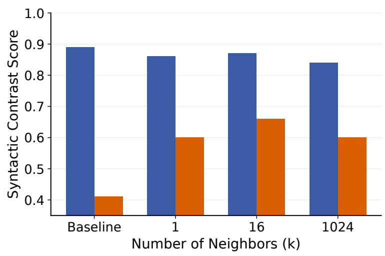
(d) Wh – token



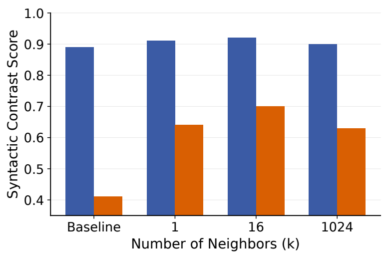
(e) Wh – phrase



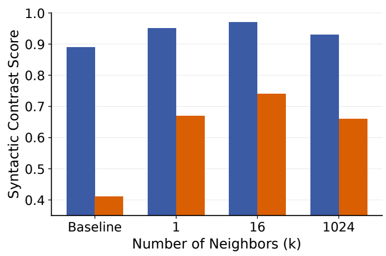
(f) Wh – sequence



(g) RC – token

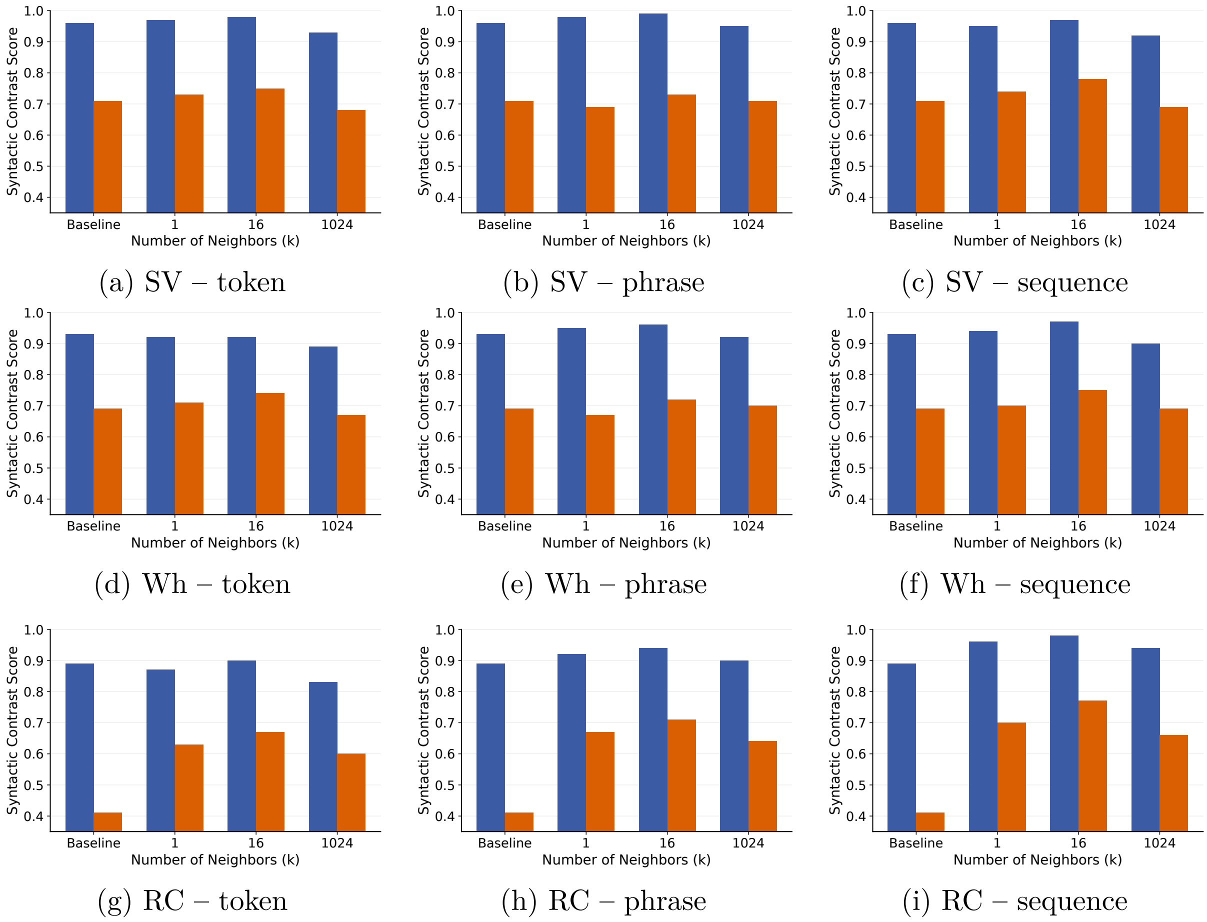


(h) RC – phrase



(i) RC – sequence

FIGURE 3.10 – Effect of number of retrieved neighbors ($k \in \{1, 16, 1024\}$) for GPT-2 XL at $\tau = 1$. Blue bars indicate high-frequency items and orange bars indicate low-frequency items; baseline results are shown for reference.

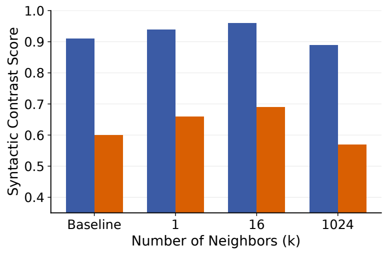


(a) SV – token (b) SV – phrase (c) SV – sequence

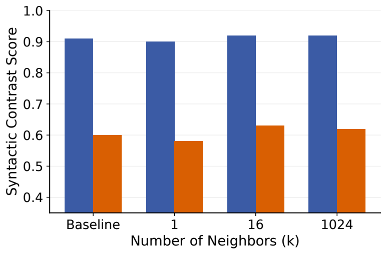
(d) Wh – token (e) Wh – phrase (f) Wh – sequence

(g) RC – token (h) RC – phrase (i) RC – sequence

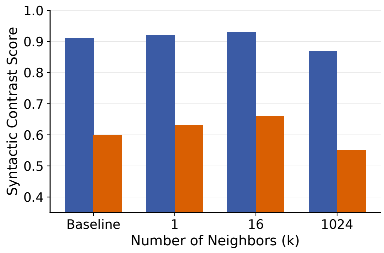
Figure 3.11 – Effect of number of retrieved neighbors ($k \in \{1, 16, 1024\}$) for GPT-2 XL at $\tau = 10$. Blue bars indicate high-frequency items and orange bars indicate low-frequency items; baseline results are shown for reference.



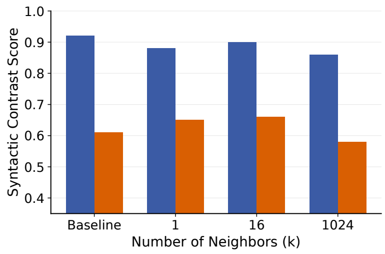
(a) SV – token



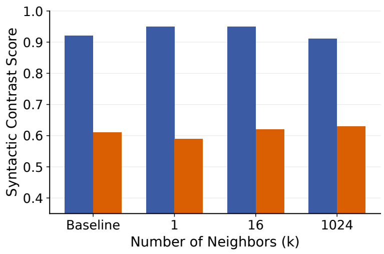
(b) SV – phrase



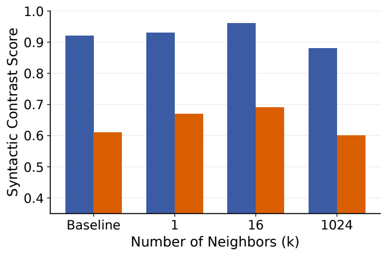
(c) SV – sequence



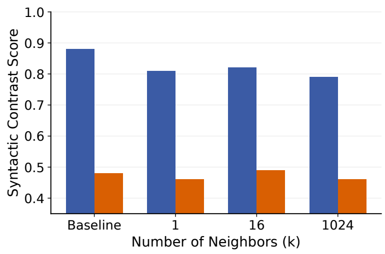
(d) Wh – token



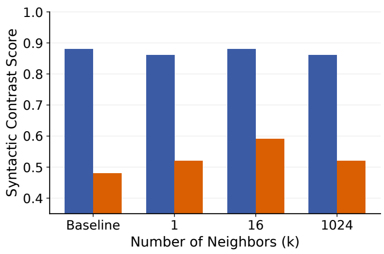
(e) Wh – phrase



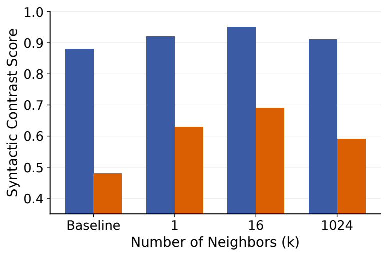
(f) Wh – sequence



(g) RC – token

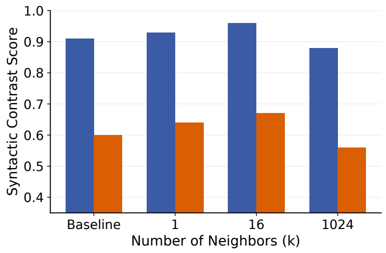


(h) RC – phrase

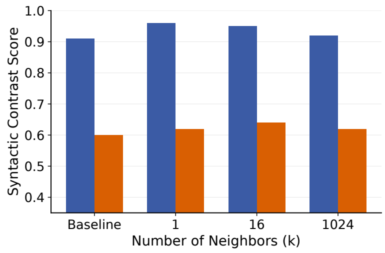


(i) RC – sequence

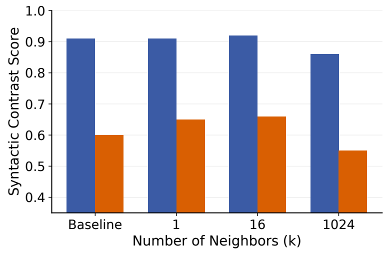
FIGURE 3.12 – Effect of number of retrieved neighbors ($k \in \{1, 16, 1024\}$) for the child-realistic model at $\tau = 1$. Blue bars indicate high-frequency items and orange bars indicate low-frequency items; baseline results are shown for reference.



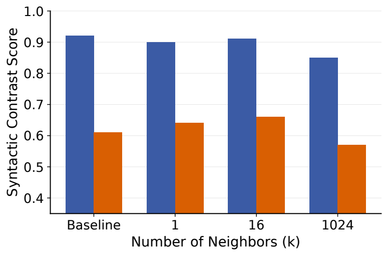
(a) SV – token



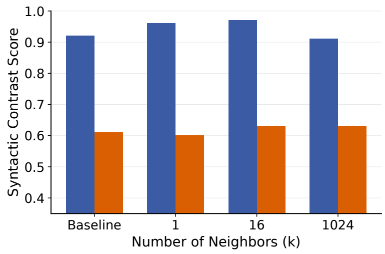
(b) SV – phrase



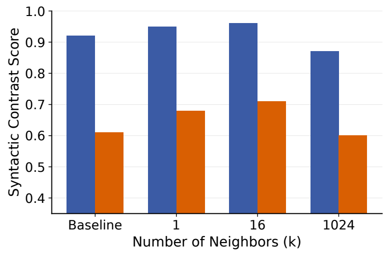
(c) SV – sequence



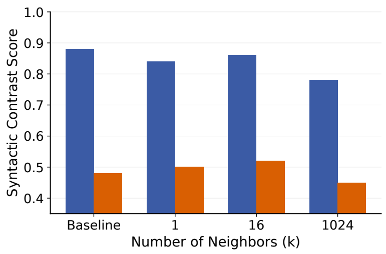
(d) Wh – token



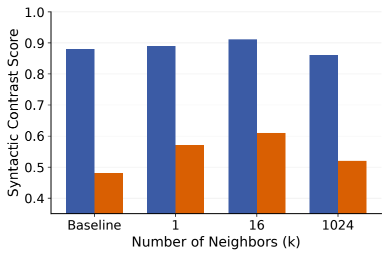
(e) Wh – phrase



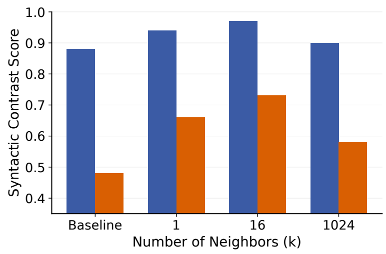
(f) Wh – sequence



(g) RC – token



(h) RC – phrase



(i) RC – sequence

FIGURE 3.13 – Effect of number of retrieved neighbors ($k \in \{1, 16, 1024\}$) for the child-realistic model at $\tau = 10$. Blue bars indicate high-frequency items and orange bars indicate low-frequency items; baseline results are shown for reference.

Chapitre 4

Modeling child-caregiver interactions: benchmarking and reinforcement-learning studies

Objectifs

Two complementary studies of how language models capture child-caregiver interaction: first, a benchmark of how state-of-the-art LLMs simulate child-directed dialogue; second, a reinforcement-learning study isolating which forms of caregiver feedback support grammar acquisition in child-like LMs.

4.1 Introduction

4.1.1 Child–Caregiver Interaction in Language Acquisition

Children encounter language in social interactions in which they are not only exposed to caregivers' utterances, but also contribute utterances of their own. Caregivers respond to these contributions, helping children express their intentions and maintain the conversation. Their responses may include requests for clarification, reformulations, or the reuse of words and structures produced by the child [Clark, 2020b ; Fusaroli, 2023]. Thus, caregiver language provides both linguistic input and responses that are contingent on children's own behavior [Masek, 2021].

This interactive context raises important questions for language acquisition. Caregiver responses may provide information about whether an utterance was understood and may offer opportunities for children to adjust their subsequent productions [Nikolaus, 2023b]. However, the availability of such responses does not, by itself, establish their role in learning. Feedback may concern the child's intended meaning, intelligibility, or participation rather than grammatical form specifically. Whether these signals provide useful information, and how a learner could exploit them, remain important questions in accounts of grammatical development [Demetras, 1986 ; Marcus, 1993]. Computational models offer a way to investigate aspects of this interaction and to test hypotheses about its possible contribution to learning.

4.1.2 Research Questions

The present chapter approaches child–caregiver interaction through two complementary questions. The first concerns the simulation of interaction: *To what extent can language models reproduce the linguistic and conversational properties of child–caregiver dialogue?* Addressing this question requires more than assessing whether individual utterances appear child-like or caregiver-like. It also requires examining how the two participants respond to one another and whether the resulting patterns remain comparable to human interaction over multiple turns. Here, language models are used as simulators, and their outputs are evaluated against observations of naturally occurring dialogue.

The second question concerns the learning effects of feedback: *Which forms of caregiver feedback can support grammatical learning in a language model?* In this case, the aim is to isolate particular signals from the broader interaction and examine their effects within a controlled learning framework. Language models are used as learners whose productions can be modified through feedback-based training. This approach allows us to compare signals that occur together in natural conversations, while distinguishing changes in generated language from changes in grammatical preferences on controlled tests.

These questions are related, but they address different aspects of computational modeling. A model may approximate observable properties of children's speech without having acquired those properties through a child-like learning process. Conversely, a simplified learning model can be used to test the effects of feedback without reproducing a complete child–caregiver conversation. We therefore consider interaction simulation and feedback-based learning as distinct uses of language models, rather than successive stages of a single system.

4.1.3 Chapter Overview

The chapter brings together two studies addressing these questions. The first develops a benchmark for evaluating language models that simulate children and caregivers, using naturally occurring conversations from CHILDES as a reference [MacWhinney, 2000]. It examines properties at the word, utterance, and dialogue levels, compares single-turn responses with extended multi-turn interactions, and assesses how providing conversational examples changes the models' behavior. Its contribution is to characterize the extent and limits of the models' ability to reproduce child–caregiver interaction.

The second study investigates learning from caregiver-derived feedback. It uses reward models trained on feedback signals extracted from natural child–caregiver conversations, rather than feedback generated by the simulated interlocutors of the first study. Small language models pretrained on caregiver input are then fine-tuned using different rewards. The study compares the effects of these rewards on the grammaticality of generated utterances and on grammatical preferences measured through minimal-pair benchmarks. This distinction is central to assessing what the feedback changes: more grammatical production does not necessarily establish the acquisition of grammatical knowledge that generalizes. The two studies thus examine, respectively, the empirical adequacy of interaction simulation and the learning effects of selected signals drawn from that interaction.

4.2 Benchmarking LLMs for mimicking child-caregiver language in interaction

4.2.1 Introduction

While LLMs show remarkable capabilities in generating human-like text and engaging in open-ended dialogues and role play in various contexts [Feng, 2024b ; Yang, 2024], their ability to simulate the specificities of child-caregiver interactions has not been systematically investigated. However, these interactions show distinct linguistic and interactive patterns and require dedicated research.

During their linguistic and communicative development, children show non-conventional (i.e., non-adult-like) patterns, such as word omissions, mispronunciations, semantic errors, and non-standard grammatical constructions [Bloom, 1993]. They also show non-conventional conversational behaviors, such as incoherence, non-responsiveness, and atypical turn-taking patterns [Ninio, 1996]. These behaviors are most apparent in the early years through primary school, although many persist into adolescence [Nippold, 2016].

Because of their still immature, non-conventional language use, children depend on caregivers to interpret and clarify their communicative intents, thus facilitating communication. Caregivers employ various *scaffolding* strategies, which offer appropriate support tailored to the child's current level of cognitive and communicative development. These include the general use of simplified language (a register named child-directed language) as well as interactive strategies such as recasting, repairing, providing follow-up, and offering feedback [Berk, 1995 ; Clark, 2020a ;

Snow, 1977 ; Nikolaus, 2023b ; Soderstrom, 2007]. This scaffolding is gradually reduced as the child becomes more proficient and ready for independent language use.

Despite growing interest in applying LLMs to specialized interactive scenarios [Feng, 2024b ; Yang, 2024], their ability to simulate child-caregiver interactions remains underexplored. There is, to the best of our knowledge, no systematic examination of whether LLMs can a) properly simulate early child-like utterances with their known non-conventional properties, b) simulate caregiver-like language and its distinctive properties known as child-directed language, and, more importantly, c) beyond mimicking child or caregiver general linguistic properties in *isolation*, simulate child–caregiver language in *interactions*, meaning that we need to simulate the fact that the caregiver's language is responsive/contingent on the child's linguistic quirks, providing tailored feedback and scaffolding. This gap is particularly significant given the potential applications of mimicking this scenario in developmental research and its applications [Zhang, 2024 ; Seo, 2024 ; Räsänen, 2024 ; Feng, 2024b].

What the current study is about

As argued above, benchmarking LLMs for effectively mimicking child-caregiver interaction is a complex, multidimensional task. The current work does not claim to provide a complete solution, but rather offers a starting point—an initial exploration into ways to approach some aspect of this task and the insights we learned that can inform future research.

In particular, it is important to note that a comprehensive benchmarking would require evaluating both the semantic content of interactions (*To what extent was the semantic content of the caregiver-like response appropriate to the child-like utterance?*) and linguistic form (*To what extent was this content linguistically framed in an age-appropriate manner?*). The current work focuses primarily on the latter, using data from spontaneous child-caregiver dialogues as a reference (CHILDES dataset)[MacWhinney, 2000].

Specifically, we quantify the extent to which two widely used LLMs (GPT-4o and Llama 3) approximate child-caregiver interactions along *structural* metrics at word, utterance, and dialogue levels. These metrics were selected based on insights from previous research [Valentini, 2023 ; Räsänen, 2024 ; French, 2024]. This previous research focused on a few isolated aspects of child-directed dialogue. Here, we aimed to provide a more synthetic view, capitalizing on insights from these studies.

Furthermore, a major novelty in this work is that we compared two benchmarking approaches: single- and multi-turn testing. In single-turn testing (the most common approach), models receive a child utterance from the CHILDES dataset and generate a caregiver-like response, or conversely, receive a caregiver's utterance and generate a child-like response. For multi-turn testing, we observe free interactions, over multiple turns, between a "child"-LLM and a "caregiver"-LLM, prompted to communicate like a child and a caregiver, respectively. The resulting dialogues are then compared, on average, to actual conversations from CHILDES dataset.

Finally, both benchmarking approaches were implemented under zero-shot and few-shot conditions with two objectives: a) for the zero-shot condition, the goal is to characterize the baseline capabilities of LLMs without specific guidance (revealing their initial biases), and b) for the few-shot condition, the goal is to measure LLMs' ability to improve (relative to the zero-shot baseline) when provided with examples and, in fact, test the sensitivity of the metrics we used to mquantify any such improvement.

4.2.2 Related Work and novelty

Research on LLMs' linguistic and interactive appropriateness to children is still in its early stages. Several studies have explored specific aspects of the interaction, but a comprehensive assessment of LLMs' capabilities in simulating child-caregiver interactions is still underexplored.

For example, VALENTINI et al. [Valentini, 2023] focused on vocabulary and showed limitations in LLMs' ability to pick simple words for a young audience, RÄSÄNEN et al. [Räsänen, 2024] trained a GPT-2 model on caregiver input, and evaluated its capacity to generate language that is similar to caregivers (child-directed language). However, they focused on simulating caregiver data alone, without accounting for the child's data, and therefore, missing the *interactive* dynamics that can influence the generation. FRENCH et al. [French, 2024] studied GPT-3.5 and Llama2's ability for linguistic alignment to the interlocutor, showing sub-optimal performance in responding appropriately to child-like utterances.

While each of these studies has contributed valuable insights to specific dimensions of the challenge, our work aims to provide a more integrated approach on, at least, two levels. First, we evaluate *both* the caregiver- and the child-like generation. We explicitly study them in interaction, not merely simulating them in isolation. Second, instead of focusing on one metric or linguistic level, we propose metrics at the word, utterance, and dialogue levels, aiming to provide a more comprehensive evaluation framework for LLMs in this specialized communicative context.

4.2.3 Method

4.2.3.1 Data

We used the CHILDES public dataset [MacWhinney, 2000] for benchmarking, focusing on 2 to 5 years of age. [1]

From this dataset, we selected 40 conversations (approximately 300 turns each) evenly distributed across the target age groups—specifically, i.e., 10 conversations at 2, 3, 4, and 5 years.

To prepare the data for analysis and generation, we restructured these conversations into utterance-response pairs. When consecutive turns came from the same speaker (which is often the case with caregivers), we decomposed these into multiple utterance-response pairs and inserted `<SILENCE>` tokens to mark the positions where the non-speaking interlocutor did not contribute. This preprocessing step preserved the temporal structure of the interactions while creating a format suitable for our analytical framework. The resulting benchmarking dataset comprised 6,600 interaction pairs containing a total of 73,300 word tokens. This corpus was further characterized by an asymmetric distribution between participant types, with 26,300 tokens produced by children and 47,000 tokens produced by caregivers.

1. This age range was selected because children younger than 2 typically do not engage in extended dialogues, while older age groups had insufficient sample sizes in the dataset to allow for robust analysis.

4.2.3.2 Models

While numerous LLMs are currently available, testing all of them across multiple experimental conditions would be impractical. Nevertheless, to examine generalizability and ensure that our results are not dependent on the idiosyncrasies of a single model architecture, we systematically compared two state-of-the-art LLMs: an open-source model **Llama 3 (8B)** (The chat-optimized version of Meta's instruction-tuned large language model, [Touvron, 2023]) and a proprietary model; **GPT-4o**, (version 2024-08), [2] the generative pre-trained transformer from OpenAI's GPT-4 family [Achiam, 2023]. We selected these models (especially the latest version of ChatGPT) as they are some of the most powerful LLMs (at least at the time this research was carried out) and thus are most likely to be capable of adapting effectively to various roles, including those of children and caregivers.

Fine-tuned model In addition to the LLMs described above, which we adapt to child-caregiver interaction via prompting, we also evaluated a smaller—and computationally more manageable—pre-trained conventional model that we directly fine-tuned on child-caregiver data. Specifically, we fine-tuned the distilled encoder-decoder BlenderBot model [Roller, 2020] on conversation data from the CHILDES corpus. [3] This approach allowed us to compare prompt-based adaptation of large, general-purpose models with traditional fine-tuning approaches targeting child-caregiver interaction patterns.

4.2.3.3 Benchmarking

Single-turn testing The single-turn testing evaluated LLM's ability to generate contextually appropriate responses to individual utterances. For each conversation in our benchmarking dataset, we implemented two complementary procedures: a) **Child-to-caregiver direction**: We extracted each child utterance and used it to prompt the LLMs to generate a caregiver-like response. The actual caregiver response from the CHILDES corpus served as the reference against which we evaluated the LLM-generated output. b) **Caregiver-to-child direction**: Conversely, we extracted each caregiver utterance and used it to prompt the LLMs to generate a child-like response. The actual child response from the corpus served as the reference for evaluation. This single-turn setting allows a direct, controlled comparison between the caregiver-LLMs' response and the caregiver's or child's actual response to the *same* utterance/prompt.

Multi-turn testing While single-turn testing enables controlled comparisons with actual responses, it cannot capture the dynamics of extended dialogue interactions. Therefore, we implemented multi-turn testing protocol to evaluate sustained conversational capabilities. To this end, we simulated complete dialogues using two distinct instances of the same LLM: one prompted to behave as a child (hereafter *child-LLM*) and another prompted to behave as a caregiver (hereafter, *caregiver-LLM*). To initiate these simulated conversations, we used the first utterance from each reference conversation in our dataset as a conversation starter. While the

2. The latest version at the moment we wrote this paper.
3. See appendix for fine-tuning details.

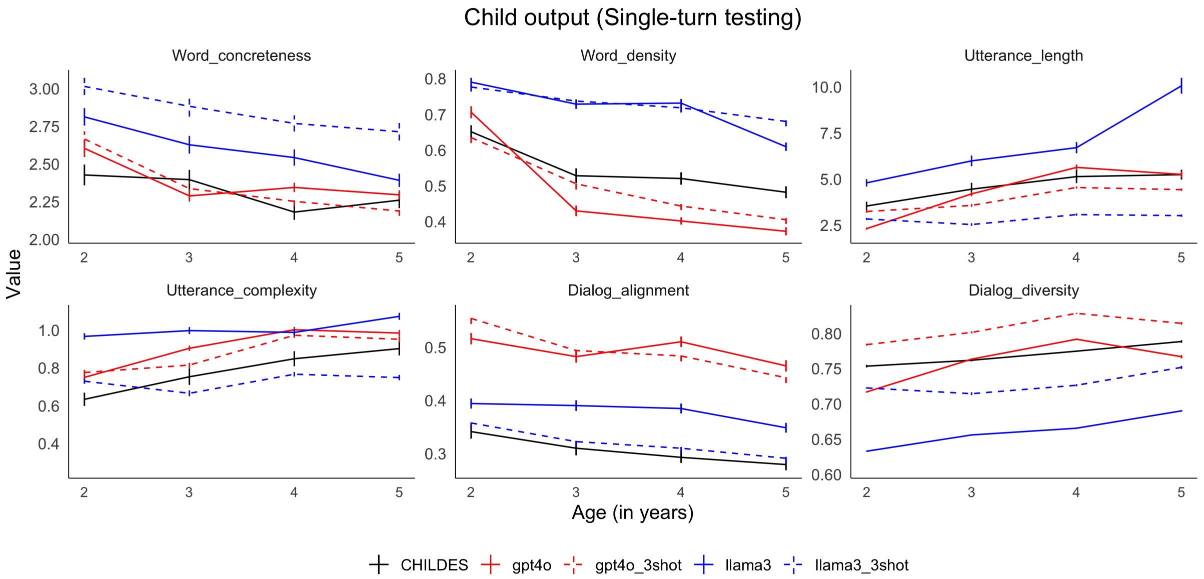


FIGURE 4.1 – Developmental trajectories of six linguistic/conversational features in children and LLMs. We compare actual human data from the child-caregiver dataset CHILDES vs. LLMs playing a child. Models include two LLMs (GPT-4o, Llama 3) in zero-shot and 3-shot settings. Points represent averages across all utterances and conversations, and ranges represent 95% confidence intervals.

outcome is not as controlled as in the single-turn case, the metrics still allow comparisons to the human reference on average, as will be clear next.

Zero vs. Few-shot settings We tested the initial built-in capabilities and biases of the LLMs in a **zero-shot** setting where LLMs received only the utterance(s) they were expected to respond to, with no additional examples from the CHILDES dataset. In addition, to test our metrics' ability to capture improvement, we tested the same models again in a **few-shot setting** where the models were given the first three child-caregiver pairs of turns in each conversation, allowing the model to observe examples of both the linguistic structure and interactive dynamics for the same participants.

In both conditions, we maintained consistent and minimal instructional prompts to avoid introducing confounding variables. For example, when prompting for caregiver responses, we used instructions such as: "*You are the parent of a X-year-old English-speaking child. Now, you are having a conversation with your child. Based on the conversation history above, give your response to the child input.*") (see the prompt templates in Appendix 4.2.7).

4.2.3.4 Metrics

We aim to provide a comprehensive benchmarking evaluating key properties at the word-, utterance-, and dialogue levels.

At the *word* level, we followed DAWSON et al. [Dawson, 2021] to quantify the *Word concreteness* using human ratings from BRYSBAERT et al. [Brysbaert, 2014]. This measure was computed

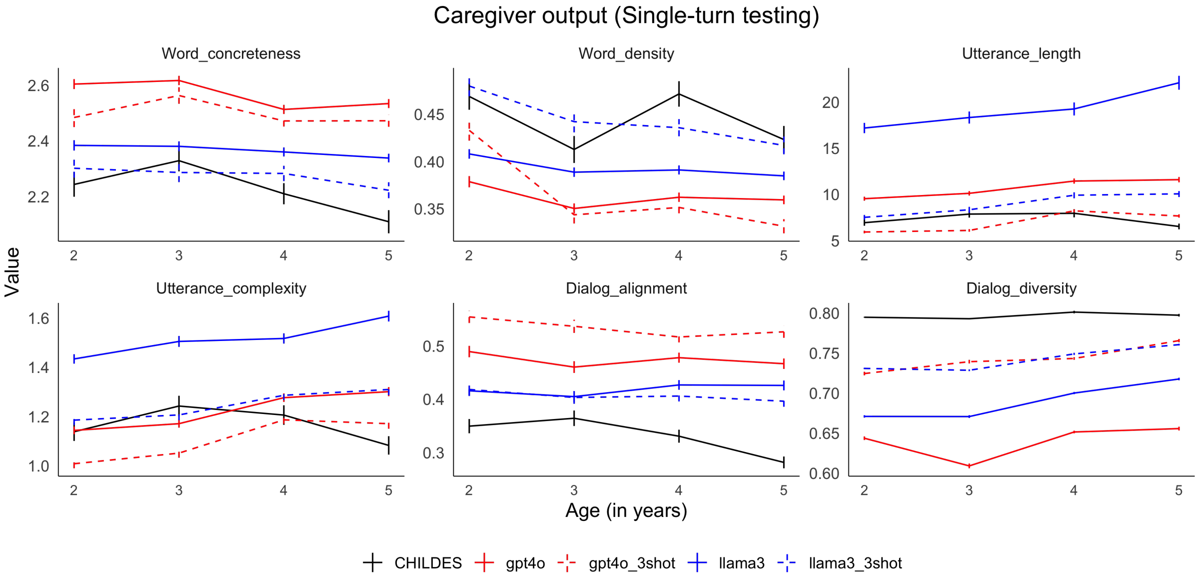


FIGURE 4.2 – Developmental trajectories of six linguistic/conversational features in caregivers and LLMs. We compare actual human data from the child-caregiver dataset CHILDES vs. LLMs playing a caregiver. Models include two LLMs (GPT-4o, Llama 3) in zero-shot and 3-shot settings. Points represent averages across all utterances and conversations, and ranges represent 95% confidence intervals.

as the average concreteness rating of all content words in each utterance. We also used *Word density*—defined as the proportion of content (vs. function) words in the utterance in each utterance. This metric reflects the information load and was calculated using the established list of function words by O'SHEA et al. [OShea, 2012].

At the **utterance** level, we adopted measures from RÄSÄNEN et al. [Räsänen, 2024] to capture structural complexity: a) *Utterance length*: the number of words per utterance; b) *Syntactic complexity*: the mean dependency tree depth for each utterance using the Spacy toolkit (v3.7) with its dependency parser based on RoBERTa transformers. [4] [Liu, 2008], where deeper trees indicate more complex structures.

Finally, at the level of the **dialogue** dynamics, we measured *Semantic alignment*, the extent to which the speaker's utterance is semantically similar to their interlocutor's (across each exchange pair) [Duran, 2019 ; French, 2024 ; Misiek, 2020b]. To this end, we used BERT sentence embedding [Reimers, 2019]. Following GUO et al. [Guo, 2023], this was calculated as the average pairwise cosine distance (1 - cosine similarity) between BERT utterance embeddings of a speaker's contributions across the entire conversation.

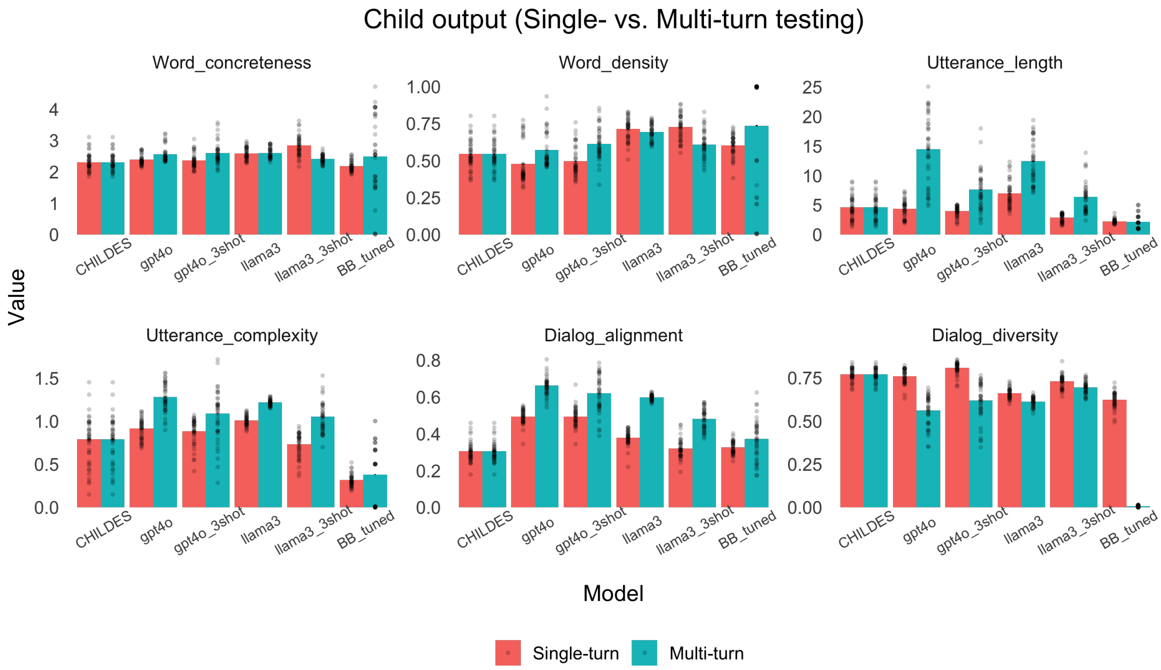


FIGURE 4.3 – Comparison of linguistic/conversational features across the single-turn and multi-turn testing for children. The bars show the averages across age groups. The points show variability across conversations (40 in total). Models include two LLMs (GPT-4o, Llama 3) in zero-shot and 3-shot settings. "BB_tuned" which stands for the BlenderBot model fine-tuned on a subset of CHILDES. Finally, for CHILDES, the data for single-turn and multi-turn is the same.

4.2.4 Results and Discussion

4.2.4.1 Single-turn testing

Figure 4.1 and Figure 4.2 show the results of single-turn testing for child-LLMs and caregiver-LLMs across age groups.

Child-LLM Our analysis reveals that child-LLMs follow, overall, the developmental patterns observed in CHILDES, even in the zero-shot setting, showing that these LLMs can simulate developmental changes in children's language without explicit guidance.

Specifically, the models capture decreasing concreteness and lexical density over time (children use more abstract words and function words as they grow older), increasing utterance length and syntactic complexity, and decreasing dialog alignment alongside increasing dialog diversity (as children become able to contribute new information instead of just repeating caregivers' inputs).

When comparing model performance, GPT-4o more accurately mimicked children than Llama 3, achieving closer alignment with CHILDES reference values across most metrics in the zero-shot condition. The introduction of few-shot examples (three interaction pairs) primarily benefited Llama 3, improving its alignment with CHILDES across several measures, though not consistently

4. The dependency parsing implementation provides a structural representation of syntactic relationships between words in an utterance.

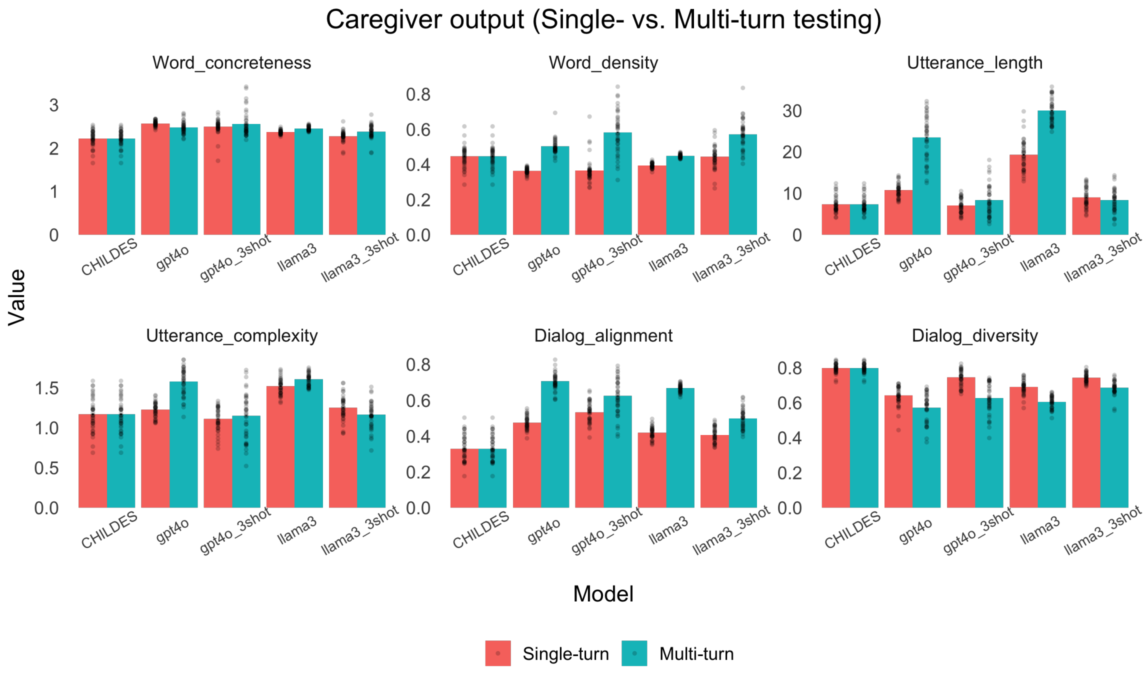


FIGURE 4.4 – Comparison of linguistic/conversational features across the single-turn and multi-turn testing for caregivers. The bars show the averages across age groups. The points show variability across conversations (40 in total). Models include two LLMs (GPT-4o, Llama 3) in zero-shot and 3-shot settings. Finally, for CHILDES, the data for single-turn and multi-turn is the same.

across all metrics. Despite these improvements, GPT-4o maintained better overall performance following few-shot prompting.

Caregiver-LLM In simulating caregiver language, zero-shot GPT-4o more closely approximated CHILDES values for utterance-level measures, while Llama 3 achieved better alignment with word-level and dialogue-level metrics. Few-shot prompting significantly improved both models' performance, with Llama 3 ultimately achieving better overall alignment with CHILDES reference values than GPT-4o after exposure to exemplars. Notably, both models continued to diverge from human data at the dialogue level, exhibiting higher semantic alignment and lower semantic diversity than actual caregivers in the CHILDES corpus. Overall, there were no clear developmental patterns in CHILDES caregivers' behaviors.

4.2.4.2 Multi-turn testing

Figure 4.3 and 4.4 show the results of multi-turn testing. For easier comparison with single-turn, we show the results side-by-side, averaged over age. Here, we made a caregiver-LLM interact with a child-LLM (using two instances of the same LLM). [5]

5. We also crossed models in child-LLM vs. caregiver-LLM (e.g., child-llama 3 interacting with caregiver-GPT-4o), but observed no noticeable changes.

Child-LLM In zero-shot settings, we observed marked differences between single-turn and multi-turn testing across multiple metrics. Most notably, utterance length, syntactic complexity, and semantic alignment all increased considerably, becoming less comparable to children in CHILDES.

After few-shot prompting, we observed minor to moderate improvements, such as reductions in utterance length [6] and (slight) decreases in both syntactic complexity and semantic alignment. [7]

Caregiver-LLM Multi-turn testing resulted in a general increase across most metrics compared to single-turn testing (with exceptions in dialog diversity, which decreased, and concreteness, which remained constant). These changes made the multi-turn behavior generally less comparable to human data in CHILDES. After the few-shot learning, we observed a significant improvement in multi-turn behavior, especially in terms of length and complexity, which became much more comparable to CHILDES. However, we observed only moderate improvement in dialog-level measures.

While LLMs successfully approximated human references for word-level and sentence-level properties under certain conditions, we identified systematic discrepancies in interactive measures for caregiver-LLMs across all experimental configurations. As illustrated in Figure 4.4, in both dialog alignment and diversity, LLMs exhibited higher alignment and lower diversity than caregiver data in CHILDES across LLM type (GPT-4o LLama 3), prompting strategy (zero- vs few-shot) and benchmarking approach (Single- vs. multi-turn).

To verify this qualitative observation statistically, we ran, for each of the two interactive measures (Dialig_alignment and Dialog_diversity), linear regressions comparing models' output to the CHILDES reference, testing all configurations: 2 LLMs x 2 promoting strategy x 2 benchmarking approach. All 16 comparison models (8 for each measure) revealed highly statistically significant differences between LLM-generated and human caregiver language, suggesting that interactive caregiver properties are potentially more challenging to mimic for LLM.

4.2.5 Conclusion

This paper presents a preliminary exploration into ways we can benchmark LLMs' ability to simulate child-caregiver interactive dynamics. While previous research has typically focused on measuring the properties of a specific structure/level (e.g., words or utterances) and/or on evaluating language from one part of the dialogue outside the interactive context (mainly focusing on the caregiver), our novel contribution is in three key aspects: a) we evaluate child-caregvier language generation in *interaction*, b) we tested a more comprehensive set of measures covering

6. Note that length could not be reduced further, even when we experimented with explicitly setting an upper bound on children's utterance in our instructions to the LLM.

7. In addition to using LLMs instructed to play a child, we also used model fine-tuning. To this end, we considered a pre-trained encoder-decoder model- Blenderbot fine-tuned on a subset of caregiver(encoder)-child(decoder) dialogues in CHILDES. We also explored an alternative approach using model fine-tuning rather than prompting. Specifically, we fine-tuned a pre-trained encoder-decoder Blenderbot model on a subset of caregiver (encoder) to child (decoder) dialogues from CHILDES. However, while showing reasonable performance in single-turn testing, this fine-tuned model was erratic and highly repetitive in the multi-turn interactions, also making it unsuitable for evaluating caregiver-LLMs.

the word, utterance, and dialog levels, and c) we compared two benchmarking approaches, evaluating the models in short and extended settings.

We put this benchmarking framework to use, comparing two powerful LLMs: GPT-4 and LLama3. Indeed, such a comparison was essential to distinguish between findings that are likely generalizable and those that are specific to a particular model. Our evaluation incorporated both single- and multi-turn testing. The former is more controlled, allowing a direct evaluation of the LLMs' response using human references, while the latter, though less controlled, allowed measuring LLMs' behavior in more extended conversations. Interestingly, we found that single-turn evaluations, while insightful, were not totally correlated with the LLMs' behavior in an extended conversation. Indeed, the multi-turn analyses showed an increased divergence of LLMs from human data, particularly in utterance-level properties and discourse dynamics (although both benchmarkings led to qualitatively similar conclusions regarding this linguistic level). This quantitative mismatch highlights the importance of dynamic testing of LLMs, since static testing alone may fail to capture the cumulative effects of sustained interactions—a consideration particularly relevant for applications involving extended communicative exchanges.

Our benchmarking effort aims not only at comparing different LLMs, but also at providing a quantitative tool that can be sensitive to incremental improvement *within* the same models. In particular, comparing zero- vs. few-shot learning, the benchmark identified areas where this intervention was more or less successful: Both GPT-4o and Llama 3 showed marked improvements in matching caregiver patterns after exposure to just three interactive examples—a finding with important implications for efficient model adaptation in resource-constrained contexts.

However, the impact was consistently less pronounced for dialogue-level properties, where models (whether in single- or multi-turn settings) continued to exhibit higher alignment and lower diversity compared to human data. This pattern suggests that while surface-level linguistic features (such as word and sentence properties) appear relatively straightforward to adapt to, capturing the interactive nature of child-caregiver communication is more challenging (see also Limitations).

In conclusion, we explored a multi-level benchmarking approach for assessing LLMs' ability to mimic child-caregiver language in interactions, and we showed its usefulness in tracking incremental improvements. We found that single-turn testing of the LLMs, as typical in most benchmarks, was not totally indicative of the LLMs' real behavior in extended conversational contexts, thus emphasizing the need for more dynamic, multi-turn testing in this line of work. Furthermore, few-shot prompting was effective in bringing the LLMs closer to caregivers' data, especially regarding word- and utterance-level properties. It was not as effective on the dialog-level properties. The LLMs exaggerated alignment and showed reduced diversity compared to CHILDES.

4.2.6 Limitations

While our explorative work has led to some initial insights, it also has several limitations and raises questions for future research.

A primary constraint concerns our selection of evaluation metrics. We relied on established measures from previous research to assess LLMs at word, sentence, and dialogue levels. For children's language production, these metrics successfully captured developmental trajectories, enabling age-specific benchmarking of LLMs (Figure 1). However, when applied to caregiver language, these same metrics failed to reveal clear developmental patterns (Figure 2), thereby limiting the precision of our assessment. This asymmetry highlights the need for developing more refined metrics that can better capture the subtle adaptations in caregiver speech across different stages of child development.

While evaluating LLMs in the single-turn scenario—comparing answers to a reference—is rather straightforward and aligns with the way LLMs are typically benchmarked, a multi-turn approach is also necessary in our context since the ultimate goal behind benchmarking is for these models to be used in an extended interactive context. However, the multi-turn approach is inherently less controlled as it involves two instances of LLMs interacting, each playing the role of an interlocutor. However, since neither instance perfectly models the interlocutor, extended interaction can amplify artifacts, leading the systems to adapt to each other's quirks rather than approximating genuine child–caregiver interaction. Thus, a multi-turn benchmarking approach can under-estimate the models' true capabilities to interact with an actual human (child or caregiver). That said, the fact that multi-turn testing was, at least qualitatively, in agreement with single-turn testing (e.g., both of them point to LLMs' ability to adapt more easily to properties of words and sentences and less easily to properties of the dialog) is a testimony to the multi-turn's potential as a valid method of evaluation in this context, though, of course, future research is needed to thoroughly investigate this question.

Finally, we reported that few-shot learning improved caregiver-LLM performance in terms of utterance length and syntactic complexity, but had less impact on alignment and diversity. While this finding was, overall, robust across configurations in our setup, it should be taken with a grain of salt, given that we did not systematically test it in a comprehensive set of experiments. Indeed, our main goal in this paper, as we stated above, was not to improve the LLMs' performance, but to demonstrate, in a simple case, the ability of the benchmarking metrics to identify gaps and track improvement. For example, it is possible that a larger, more systematic exploration of the prompting strategies (which is computationally expensive, and thus, could not be done here in combination with the other experiments we did) could yield improvements across the dialog dimensions as well.

4.2.7 Ethics statement

All data used in this study is already publicly available. This work focuses on model benchmarking and improvement using offline child-caregiver data and internal simulations, aiming to advance fundamental research in this area. We do not consider this testing sufficient for deployment; any future real-life applications should undergo rigorous validation in child-safe environments with appropriate human oversight, such as by teachers or parents.

Acknowledgments

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Appendix A: Prompt templates

We provide prompt templates in different conditions. These are templates for the multi-turn testing. The templates for the single-turn testing are the same, except there is no conversation history.

The zero-shot prompt template for the caregiver [**Conversation history** ADULT: <Utterance>, CHI: <Utterance>...] You are the parent of a <Month>-month-old English-speaking child. Now, you are having a conversation with your child. `<SILENCE>` indicates silence in the previous turn; `<UNINTELLIGIBLE>` indicates unintelligible speech. Based on the given conversation history above, give your response to parent input as ADULT. Do not output the speaker label.

The zero-shot prompt template for the child [**Conversation history** CHI: <Utterance>, ADULT: <Utterance>...] You are a <Month>-month-old English-speaking child. Now, you are having a conversation with your parent. `<SILENCE>` indicates silence in the previous turn; `<UNINTELLIGIBLE>` indicates unintelligible speech. Based on the given conversation history above, give your response to parent input as CHI. Do not output the speaker label.

The few-shot prompt template for the caregiver [**Conversation history** ADULT: <Utterance>, CHI: <Utterance>...] You are the parent of a <Month>-month-old English-speaking child. Now, you are having a conversation with your child. `<SILENCE>` indicates silence in the previous turn; `<UNINTELLIGIBLE>` indicates unintelligible speech. Ensure your response is no longer than 50 words regardless of the prompt. Here are some example interactions: CHI: <Utterance>, ADULT: <Utterance>... Follow the example interactions. Based on the given conversation history above, give your response to parent input as ADULT. Do not output the speaker label.

The few-shot prompt template for the child [**Conversation history** CHI: <Utterance>, ADULT: <Utterance>...] You are a <Month>-month-old English-speaking child. Now, you are having a conversation with your parent. `<SILENCE>` indicates silence in the previous turn; `<UNINTELLIGIBLE>` indicates unintelligible speech. Ensure your response is no longer than 6 words regardless of the prompt. Here are some example interactions: CHI: <Utterance>, ADULT: <Utterance>... Follow the example interactions. Based on the given conversation history above, give your response to parent input as CHI. Do not output the speaker label.

Appendix B: Finetuning details

We fine-tuned the Blenderbot model, a 400M parameter architecture comprising: a retriever for dialogue history; a Seq2Seq generator with 2 encoder layers, 24 decoder layers, 2560-dimensional embeddings, and 32 attention heads; retrieve-and-refine architectures combining dialogue and knowledge retrieval. The parameters were optimized through preliminary experiments, resulting in a batch size of 16 and a learning rate of 0.0001 maintained over 4 epochs. The fine-tuning data consisted of 2.6M interaction pairs with 15.8M total word tokens (4.4M child tokens and 11.4M adult tokens) restructured from CHILDES dataset [MacWhinney, 2000].

Appendix C: Example generations

We present sample generations in Table 4.1. It shows child-caregiver conversations (CHILDES corpus) and GPT-4o generations in a multi-turn setting where the child and caregiver are two instances of GPT-4o. All conversations start from the same child turn (here, all conversations start from a child outputting "silence"). To extract features, we preprocessed all samples by removing non-verbal behavioral annotations (e.g., *giggles*) and lower-casing words to focus exclusively on verbal content.

TABLE 4.1 – Comparison of real and LLM-generated child-caregiver dialogue exchanges. The LLM generation here is from a multi-turn setting, where two instances of GPT-4o (i.e., a child-LLM and a caregiver-LLM) interact. The table shows three dialogue samples: (1) a real interaction from the CHILDES corpus, (2) a zero-shot generation by GPT-4o given only the initial turn, and (3) a few-shot generation by GPT-4o after being shown three previous examples in the child-caregiver interaction.

Source	**Role**	**Utterance**
CHILDES (Human)	C	*SILENCE*
	A	wanna play catch
	C	let's play catch
	A	okay almost
	C	yeah
	A	try it again put your hands in like this
	C	almost
	A	you wanna hit it
GPT-4o (Zero-shot)	C	*SILENCE*
	A	I see you're still thinking about cherries. It's okay to feel upset. Would you like a hug while we figure out something else yummy to eat?
	C	*SILENCE*
	A	I know you're feeling quiet right now. I'm here when you're ready. Would you like to hold the banana and see if it feels fun to peel?
	C	Banana! Peel it?
	A	Yes, let's peel the banana! Can you help me pull the peel down? Then we can eat it together. Yum!
	C	Banana! Peel! Eat!
	A	Yes! Let's peel the banana together. Peel, peel! Now take a bite—yum! Is it tasty?
GPT-4o (Few-shot)	C	*SILENCE*
	A	Do you love your big red truck?
	C	Big red truck! My truck!
	A	Is it your big red truck? Where will you drive it?
	C	Drive to park! Big red truck!
	A	Are we driving your big red truck to the park? That sounds so fun!
	C	My truck drive park!
	A	Yes, your big red truck! We can drive it to the park!

Notes: A = Adult, C = Child. All model-generated responses were obtained using default sampling parameters (temperature = 1.0). *SILENCE* marks an empty turn. Human dialogues are from the CHILDES corpus [MacWhinney, 2000].

4.3 Which forms of caregiver feedback support grammar learning?

4.3.1 Introduction

Children do not learn language from exposure to input alone, but also through social interaction in which caregivers respond to their produced utterances [Fusaroli, 2023 ; Masek, 2021 ; Clark, 2020b]. A long-standing question in language acquisition concerns the role of such feedback in shaping grammatical development, especially if caregiver responses can help the learner know when their productions were well- or ill-formed [see review in Sokolov, 1994].

Although caregivers rarely provide explicit correction of grammatical errors [Brown, 1970], researchers have proposed that they nevertheless provide global, noisy feedback from which children can infer whether their utterances were likely grammatical or ungrammatical [e.g., Hirsh-Pasek, 1984 ; Demetras, 1986 ; Sokolov, 1994]. In this view, the child can be understood as learning in a reinforcement-like fashion, using the valence of the feedback as positive or negative rewards that guide which productions to maintain or adjust in future interactions [**schoneberger2010three** ; e.g., see Whitehurst, 1988 ; Nikolaus, 2023a].

The developmental literature has documented several forms of caregiver response that can be interpreted, within a reinforcement-learning framework, as valenced feedback. The most direct examples are *communicative feedback* cues, namely a) acknowledgments (e.g., "okay", "yeah"), which signal communicative success and thus increase the likelihood that the child's sentence was well-formed, and b) clarification requests (e.g., "huh?", "what?"), which signal communicative failure or misunderstanding and thus increase the likelihood that the sentence was ill-formed [Clark, 2020b ; Demetras, 1986 ; Newport, 1977 ; Saxton, 2005c ; Nikolaus, 2023c ; Nikolaus, 2022].

A second relevant class is *semantic contingency*, or the extent to which caregiver responses remain responsive to the child's topic and intended meaning [**agrawal2026scaffolding** ; Hoff-Ginsberg, 1987 ; Che, 2018 ; Hirsh-Pasek, 2015 ; Agrawal, 2024]. The degree of this contingency captures the feedback valence: low contingency in the caregiver response signals semantic misalignment and thus suggests that the child's utterance may be ill-formed.

There are other forms of caregiver feedback that also provide such valenced signals, even if they have not traditionally been associated with this learning framework. Perhaps most directly relevant to grammar learning is *structural alignment* [Fusaroli, 2023 ; Dale, 2006 ; Fernandez, 2014 ; Misiek, 2020a], whereby caregivers respond by reusing the child's syntactic frame while potentially introducing different lexical content (e.g., child: "I eat" (Pronoun-Verb) → caregiver: "Mommy eats too"). Here, the feedback valence is reflected in whether such alignment occurs, signaling support for the child's use of that syntactic frame.

A final type of feedback is *affective feedback*, such as whether caregivers respond with positive affect (e.g., "that's amazing!") versus a more emotionally neutral response [Rivero, 2023].

These cues are far more available to children than explicit grammatical corrections, making their role in language learning much more plausible [Demetras, 1986 ; Hirsh-Pasek, 1984]. Yet, they provide only indirect and noisy learning signals. First, they are not always directed at grammar specifically. Second, even in cases when grammar is the main target, the signal is global (i.e., utterance-level) and does not identify the specific error within the utterance. This makes *credit assignment* a central challenge [Marcus, 1993 ; Pinker, 1989 ; Saxton, 1997], and also makes interactive learning from caregiver feedback an interesting *empirical* case.

Nevertheless, little is known about how these different feedback types contribute to children's grammatical learning. While corpus studies allow a quantitative account of caregivers' feedback behavior, *the causal effect* of each type of feedback or reward on learning remains difficult to establish. A common methodological issue facing previous attempts [e.g., see discussion in Saxton, 2005a] is that different types of feedback are provided to the same child, making it difficult—and ethically problematic—to isolate their effect over the developmental timescales on which language is learned.

Recent computational work has begun to revisit these questions by using LMs as controlled learning environments for testing hypotheses about language learning, thereby combining the targeted precision of experimental methods with the ecological validity of corpus studies [e.g., Misra, 2024]. Particularly relevant to our question are studies that used Reinforcement Learning to investigate LMs' interactive learning strategies [Ma, 2024 ; Mayer Martins, 2025 ; Zhao, 2023 ; Padovani, 2025a ; Xu, 2025 ; Nikolaus, 2021b].

Recent work has applied this approach to *natural* child–caregiver interactions, using Reward Models trained from caregivers' actual feedback data [Nikolaus, 2026]. Language models are then fine-tuned using techniques inspired by Reinforcement Learning from Human Feedback [RLHF, Ouyang, 2022]. This work showed that caregiver-like clarification requests on model generation can signal negative evidence. Learning from this signal improved grammaticality.

The current study

Existing RLHF-based studies of naturally occurring caregiver feedback have focused primarily on clarification requests, leaving open whether other forms of social feedback have similar, weaker, or even negative effects on grammatical learning. The present study addresses this gap by comparing several theoretically motivated feedback types within the same controlled learning framework.

4.3.2 Methods

Our method uses a two-stage learning setup designed to separate learning from linguistic exposure from learning driven by caregiver-feedback rewards. In a first step, a small causal language model is pretrained on child-directed language (excluding children's production), providing an *input-only* baseline that quantifies learning gains from linguistic exposure. In a second step, the model is fine-tuned via RL, using a reward model trained on the target caregiver's feedback valence.

Pre-training baselines We use a customized version of GPT-2 (2 layers, hidden size 512, 8 attention heads), pre-trained on CHILDES [MacWhinney, 2000], using three pre-training scales (0.1M, 1M, and 10M words, see Appendix 4).

Empirical Reward Models For each feedback type, we trained reward models to capture

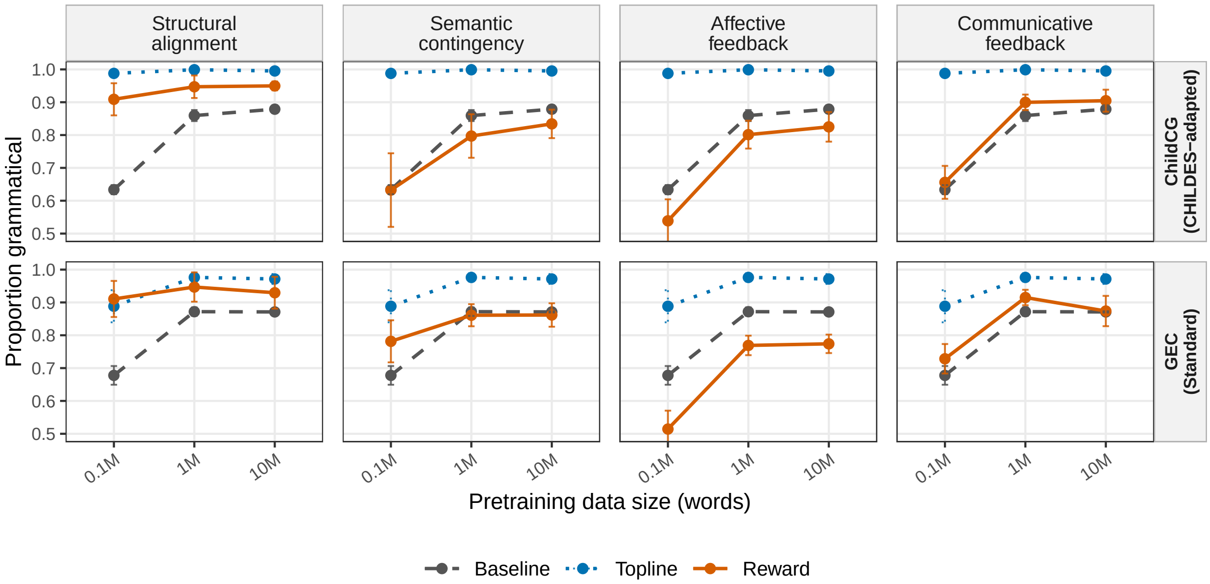

FIGURE 4.5 – Generated-utterance grammaticality across pretraining scales for each reward. The top row shows results from the CHILDES-adapted ChildCG classifier, while the bottom row shows results using a standard GEC-based metric. Within each panel, baseline and topline trajectories are repeated to make the reward trajectory directly comparable. Means are computed over training seeds units after averaging over generation seeds. Error bars show ±1 standard deviation across training seeds.

naturally occurring patterns of caregiver feedback in child–caregiver interactions (using CHILDES dataset). Each reward model is trained to map children's utterances to a scalar, representing the valence of the target feedback (see Appendix 4). The valence values themselves were annotated prior to reward-model training using a range of automatic annotation tools (see below, and Appendix 3).

Topline Reward Model: we use an idealized reward that directly targets grammatical errors. This topline is not intended to produce empirically realistic results, but rather to serve as an upper bound on what grammatical learning is possible *in principle* (see details in Appendix 4).

RL fine-tuning For each feedback/reward, the pre-trained models were fine-tuned using the corresponding reward models, using Proximal Policy Optimization [PPO; Schulman, 2017]. We included several safeguards to mitigate potential language drift (see Appendix 5).

Training seeds and variability To account for stochastic variability, we used, for each training scale, a fully crossed $3 \times 3 \times 3$ design over pretraining seeds, fine-tuning seeds, and generation seeds.

Annotation of the reward signals We considered the four classes of caregiver feedback mentioned in the introduction: communicative feedback, semantic contingency, structural alignment and affective feedback. For each type of feedback, we annotated the CHILDES child-caregiver conversations for the corresponding valence (See Appendix 3).

4.3.2.1 Evaluation

We use two complementary evaluations that capture different aspects of grammatical behavior. **Minimal-pair benchmarks**: we use Blimp [Warstadt, 2020] and its CHILDES-adapted version Zorro [Huebner, 2021b]. They test whether a model assigns higher probability to a grammatical sentence than to a minimally differing ungrammatical one.
Grammaticality of generated utterances Free generations were scored using Child Conversational Grammaticality [ChildCG, Nikolaus, 2024], a model adapted to CHILDES and its conversational context. In addition, we used a measure based on an off-the-shelf Grammatical Error Correction model [e.g., Rothe, 2021]. For each run and generation seed, we sampled up to 10,000 utterances using the Beginning-Of-Sentence (BOS) token as a prompt.

Both evaluation methods are broadly connected to child-language methodologies for evaluating grammatical development, which have included both controlled forced-choice tasks and analyses of children's spontaneous and elicited productions [Ambridge, 2011].

4.3.3 Results

First, we compared the baseline models, trained only on linguistic input, with the same models further fine-tuned using the grammar topline reward model (Figure 4.6 in Appendix). The topline improves both minimal-pair benchmark performance and the grammaticality of generation, showing the current framework can, in principle, lead to *grammatical* learning above and beyond pre-training.

Moving to the empirical reward models, we found no consistent improvement on the minimal-pair benchmarks (Figure 4.7 in Appendix). Across Zorro and BLiMP, none of the reward types reliably improved over the baseline. This echoes a broader pattern in recent studies showing that interactive strategies struggle to improve performance on these benchmarks (see also Limitations). In contrast, the generation results showed measurable effects on grammaticality (Figure 4.5). Across both ChildCG and GEC, the overall trends were largely consistent. Structural alignment produced the strongest and most consistent improvement across scales and metrics. Communicative feedback showed an overall positive effect, especially under ChildCG, but the gains were smaller. In contrast, semantic contingency and affective feedback generally reduced grammaticality below the baseline.

Regarding the effect of pre-training size, the low-resource condition at 0.1M was visibly noisier, with larger variability across runs, excessively high non-word rates (Figure 4.8 in Appendix, bottom row), and overall low grammaticality scores. The effects became much more consistent from 1M words onward, suggesting that a minimum amount of pre-training data is needed before reward-specific effects can emerge robustly.

We tested these qualitative patterns statistically using a mixed-effect logistic model. Predictors included reward type, pretraining scale, and their interaction. We additionally added the following predictors as controls a) utterance length, since longer utterances may naturally provide more opportunities for errors, and b) rate of non-words generated, since out-of-dictionary forms may be penalized by grammaticality classifiers for reasons that are not strictly grammatical. Finally,

Reward family	1M	10M
Structural alignment	$+.077^{***}$	$+.060^{***}$
Semantic contingency	$-.092^{***}$	$-.061^{***}$
Affective feedback	$-.038^{*}$	$-.031^{*}$
Communicative feedback	$+.033^{*}$	$+.024^{\dagger}$
Acknowledgement	$+.017$	$+.017$
Clarification request	$+.050^{***}$	$+.031^{*}$

TABLE 4.2 – Model-estimated change in grammaticality probability relative to baseline, using the ChildCG metric at the 1M and 10M pretraining scales. $\dagger p < .10$, $^{*}p < .05$, $^{**}p < .01$, $^{***}p < .001$.

we included a random intercept for generation run to account for the fact that sampled utterances from the same seeds are not independent.

Table 4.2 reports model-estimated changes in grammaticality probability relative to the baseline, using the more robust pretraining scales (1M and 10M). Mirroring the pattern in Figure 4.5, structural alignment showed the clearest improvement. In contrast, semantic contingency and affective feedback both reduced grammaticality. Communicative feedback improved grammaticality at 1M and showed a marginally significant positive effect at 10M. When communicative feedback was split into its two components, only clarification requests showed a significant effect, whereas the acknowledgment-based reward model did not.

4.3.4 Discussion and Conclusions

We investigated which forms of caregiver feedback support LMs' grammaticality within an RLHF-inspired framework. We isolated and compared major feedback types, providing insight into their specific effects and relative contributions.

The main finding was that Structural Alignment systematically improves grammaticality. While previous work found correlational evidence to this link [Fusaroli, 2023], the current result demonstrates, at scale, a novel and plausible *mechanistic* account: Caregiver's structural alignment can support grammar learning *because* this reinforces the child's use of structures that are grammatical—caregivers naturally tend to (re-)use grammatical sequences more often than ungrammatical ones. Communicative Feedback also improved grammaticality, though to a lesser extent. The improvement is consistent with prior work, especially on clarification requests [Nikolaus, 2026 ; Clark, 2020b].

As for Semantic Contingency and Affective Feedback, they *reduced* grammaticality. We did not anticipate this outcome; however, it can be understood in the broader context of development. Based on the supplementary analyses in Figure 4.8, and models' generation samples in Table 4.3 (Appendix), Affective Feedback increased utterance length, suggesting a role in eliciting longer vocalization [see also Goldstein, 2003]. Semantic Contingency increased the use of content words and preserved greater lexical diversity, in line with findings in lexical development [Che, 2018].

To conclude, different types of feedback push generation in different directions: some improve grammaticality, while others encourage other properties of language learning and use. Future work should adopt a broader framework, investigating how these feedback types may support *complementary* aspects of language learning beyond grammar.

Limitations

A limitation of our study is that the effects of the empirical rewards were observed in model generation, but not in minimal-pair benchmarks. This echoes a broader pattern in recent studies showing that interactive strategies generally struggle to improve performance on minimal-pair benchmarks of grammar such as BLIMP and Zorro [Zhao, 2023 ; Padovani, 2025a ; Nikolaus, 2026 ; Stöpler, 2025].

This raises an important theoretical question: To what extent does grammaticality in generation reflect grammatical knowledge? This remains a debated question in child language research [Ambridge, 2011 ; Chater, 2018] ; and we do not aim to address this general question here.

In the specific case of the present study, however, it is cautious to interpret the generation-based results more narrowly: they test how reward fine-tuning changes the model's productive behavior; they do not directly test whether the model has acquired grammatical knowledge that can generalize. Note that even under this conservative interpretation, the generation-based results remain informative for theories of language development. Unlike traditional corpus-level correlations, the rewards here were used as interventions within a computational learning system, allowing us to test whether optimizing for these signals *causally* shifts the model's production policy in grammar-relevant directions.

Nevertheless, a model that becomes more grammatical in free generation does not necessarily mean it has learned grammatical regularities that generalize beyond the specific utterance types favored during fine-tuning. Minimal-pair benchmarks—though not perfect [Martínez, 2023]—are useful precisely because they intend to test this stronger criterion for learning. Our null results on BLiMP/Zorro (Figure 4.7) therefore matter. They suggest that the empirical rewards did not produce grammatical generalization—at least, not of the kind captured by these benchmarks.

That said, while this conclusion may be partly correct, it would be premature to draw it too strongly here without further research. Existing minimal-pair benchmarks do not exhaust the space of possible grammatical generalizations. In particular, even child-adapted benchmarks such as Zorro primarily target classic grammatical phenomena, but do not directly target some of the errors that are especially common in child language *production* and may therefore also be reproduced in our generated data—including missing subjects, objects, auxiliaries, or determiners, among others [Saxton, 2005b ; Hiller, 2016 ; Nikolaus, 2024]. This is important because, in an RL fine-tuning setting, errors that are sampled in the model's productions are the ones that receive direct corrective pressure from the reward signal.

Another concern is that reward fine-tuning may narrow the model's production distribution, reducing diversity and potentially harming generalization. However, while this may be partly true, some narrowing may still be compatible with grammatical learning, and perhaps even expected if the model moves away from noisy or ill-formed productions. For instance, in our analyses, the grammar topline was relatively the least lexically diverse condition (Figure 4.8, Lexical Entropy, Appendix), yet it produced the clearest gains on the minimal-pair benchmarks (Figure 4.6, Appendix). Thus, reduced lexical diversity does not in itself prevent grammatical generalization. The more central question is whether reward-induced changes target grammatical phenomena that can generalize beyond the sampled utterances themselves.

Testing this stronger learning claim, thus, remains an important direction for future work. It requires conducting large scale corpus exploration of the grammatical errors found in both child and model productions, and to use these bottom-up insights to build targeted tests around the relevant phenomena. This would help determine whether the reward-induced gains observed here reflect merely local shifts in surface production patterns or more abstract grammatical generalizations.

Ethical considerations

This work uses existing child-caregiver corpora and does not involve new data collection. Because the data involve children, we treat the corpora as sensitive and report only aggregate model-level results, without attempting to identify individual children, caregivers, or families. The proposed modeling framework is intended as a tool for studying hypotheses about language development, not for evaluating individual children or caregivers.

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Appendix 1: Supplementary Analyses

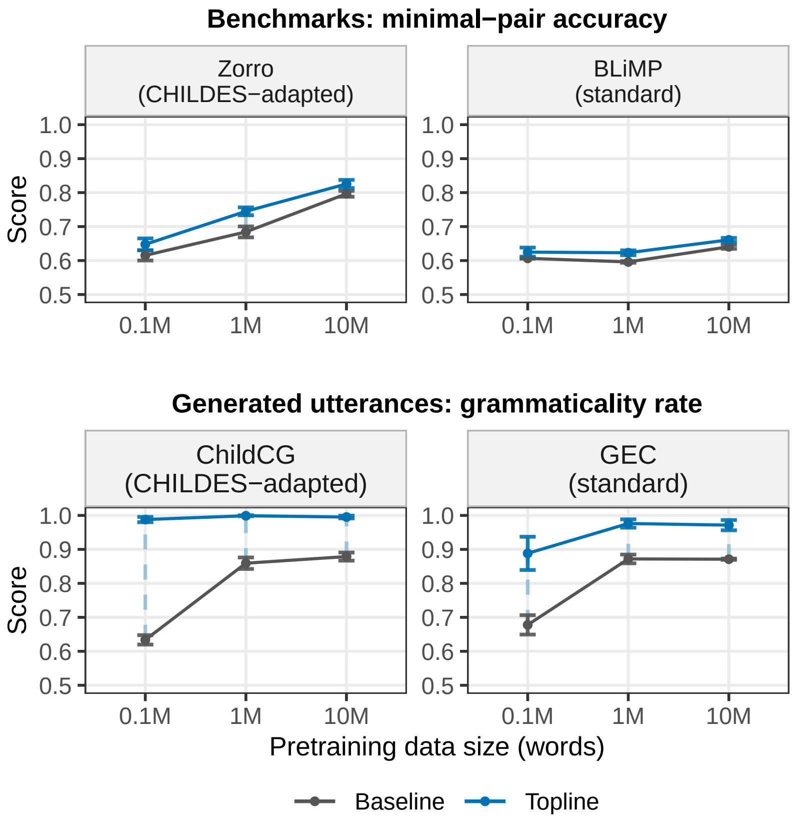

FIGURE 4.6 – Baseline and Topline performance across pretraining scales. The top row reports minimal-pair benchmark accuracy for Zorro and BLiMP. The bottom row reports grammaticality of generated utterances, using the ChildCG or GEC-based metric. Points show means across trainings seeds (after averaging over generation seeds). Error bars show ±1 standard deviation across these independent units.

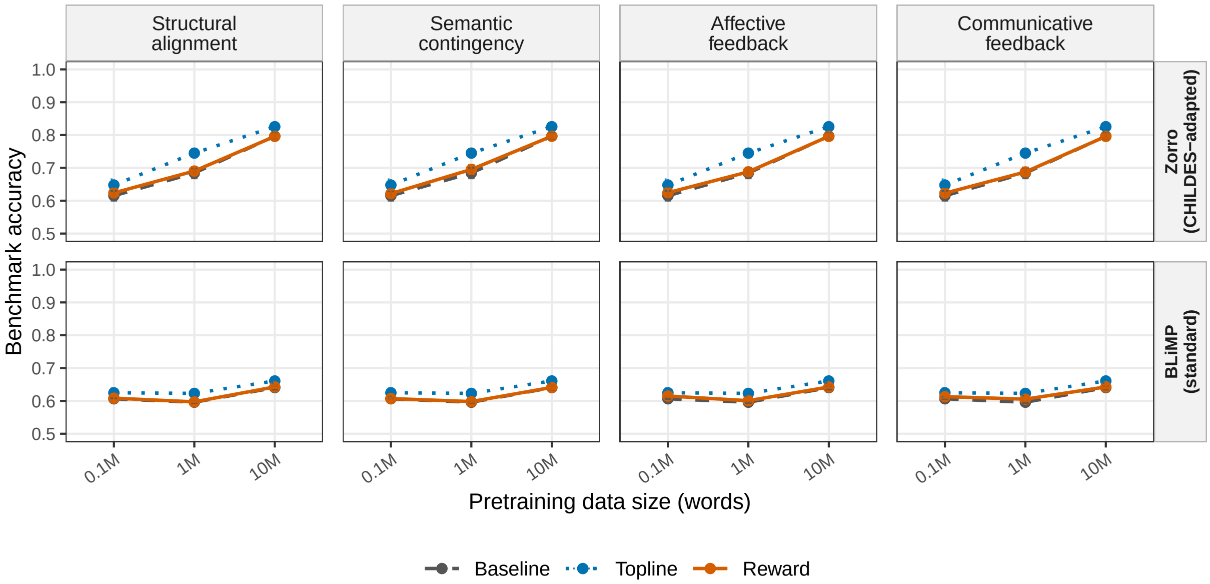

FIGURE 4.7 – Benchmark performance across pretraining scales for each reward family. Rows show Zorro and BLiMP minimal-pair accuracy; columns show the four empirical reward families. Within each panel, the baseline and topline trajectories are repeated to make the reward trajectory directly comparable at each scale. Means are computed over training seeds, after averaging over generation seeds. Error bars show ±1 standard deviation across independent units.

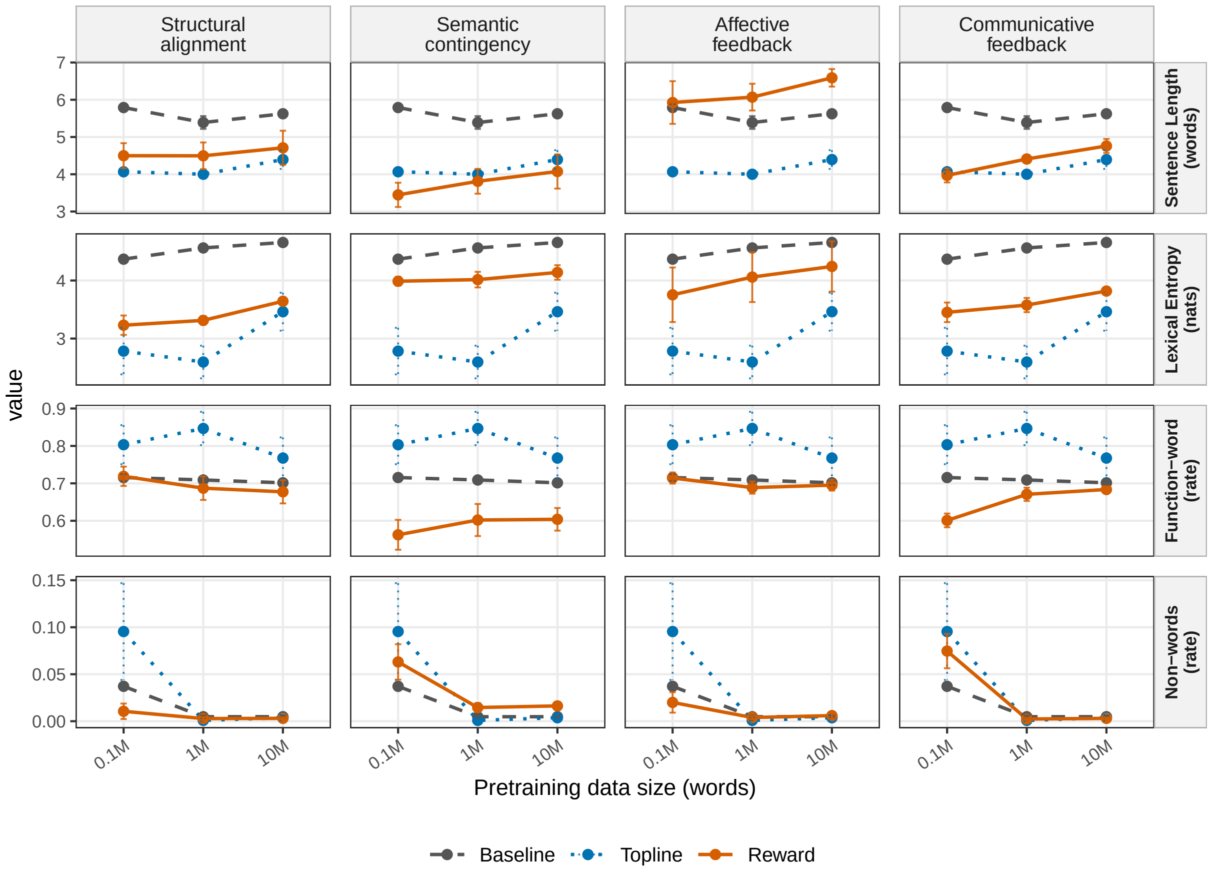

Figure 4.8 – Properties of generated utterances: Sentence length, pooled lexical entropy, function-word density, and rate of non-words. Points show means for each reward across training seeds, with error bars indicating ±1 standard deviation across these seeds. Within each panel, baseline and topline trajectories are repeated to make the reward trajectory directly comparable at each scale.

Appendix 2: Examples of Model-Generated Utterances

Reward type		ChildCG = 1 (grammatical)	ChildCG = 0 (ungrammatical)
Baseline		*you want to look at some more pictures?* *oh, is that what it looks like?* *it is a big boy now isn't it?* *have you not any other little boys?*	*this piece of cake?* *because you can see mummy's poorly sick.* *the dog is saying.* *he is the name of the little girl.*
Topline		*what are you doing?* *there is a horse.* *that is a girl.* *are you going back?* *that is a chair.* *what is that one?* *i will put it there.* *what did we do with the barn?*	*why is that side?*
Structural Alignment		*so that is a duck.* *he is a turtle.* *you are a good boy.* *i am a monkey.*	*how many trucks in the sky.* *come down the street.* *we are a good boy.* *can't find it.*
Semantic Contingency		*yes that is a doggie.* *well it is a boat.* *there is an elephant.* *yes a banana puppy.*	*where is baby sheep?* *what is big truck?* *he a lion.* *a big flower red.*
Affective feedback		*and then i will put it in it.* *i will put it over here if i put it in there.* *we have to wash her hair first okay?* *we will shall we put it all together?*	*if i do it and then we can do it and put it in?* *now i will have to clean this with your room won't you?* *want mama do it?* *because your teeth i am watching this picture aren't they?*
Communicative feedback	Acknowledgement	*they are called flowers.* *that is a gold straw.* *and that is the baby.* *you have a pink one on there.*	*have to use your spoon and pretend.* *you will use them ones with this bits.* *all day and then the is warm.* *but two grey stars.*
	Clarification request	*i don't know.* *can you see it?* *can't you?* *mhm, it is.*	*look at that isn't it?* *look it, there is it?* *see you shouldn't?* *there doesn't.*

TABLE 4.3 – Examples of utterances generated by RL-fine-tuned models at the 10M-word scale (generation was qualitatively similar at the 1M-word scale). Examples are organized by reward types. Unshaded cells show utterances that were classified as grammatical by ChildCG, while shaded cells show utterances classified as ungrammatical.

Appendix 3: Automatic labeling of rewards

We automatically annotated caregiver feedback signals in child-caregiver conversations and used these annotations to supervise reward model training (see Appendix 4).

Communicative feedback

Clarification requests Clarification requests were identified using a DeBERTa-v3-xsmall classifier fine-tuned on manually annotated caregiver responses in CHILDES, following prior work [Nikolaus, 2026].

Acknowledgements To identify caregiver acknowledgements, we relied on the feature-based algorithm described in Nikolaus et al. [Nikolaus, 2023c], which was specifically designed for CHILDES data and combined common keywords (e.g., “okay”, “alright”, and “yeah”) as well as repetition ratios capturing repetition-based acknowledgements (e.g., “It isn’t very nice, is it?” – “It isn’t.”).

Structural Alignment

Following previous research [Dale, 2006 ; Fernandez, 2014], a child structure was defined as a POS bigram. We used spaCy (`en_core_web_sm`)[8] to tag each child utterance and caregiver response for part-of-speech (POS), excluding punctuation and empty tokens. Caregiver alignment was computed as a normalized score of bigram re-use, corresponding to the size of the intersection between the child and caregiver POS-bigram sets, divided by the size of the larger set.

Semantic Contingency

Following previous research [e.g., Fusaroli, 2023 ; Foushee, 2022], semantic contingency was computed using sentence embedding similarity. Here, we used `all-MiniLM-L6-v2`[9] to encode the child utterance and caregiver response. We computed the cosine similarity between the two embeddings, and clipped the resulting score to the $[0, 1]$ range.

Affective feedback

We annotated affective properties of caregiver responses using `roberta-base-go_emotions`.[10] The model was applied to adult responses only. For each caregiver utterance, the classifier returned a probability distribution over several labels. We used the probabilities for emotions related to supportiveness, approval, and warmth.

8. https://huggingface.co/spacy/en_core_web_sm
9. https://huggingface.co/sentence-transformers/all-MiniLM-L6-v2
10. https://huggingface.co/SamLowe/roberta-base-go_emotions

Appendix 4: Baseline and Reward model training

Baseline models

For the input-based baselines, we trained a small GPT-2-style causal language model on caregiver utterances only. The model used 2 hidden layers, 8 attention heads, and a hidden size of 512. We trained a word-level tokenizer on the training data, retaining words that appeared at least twice and capping the vocabulary at 5,000 types. Models were optimized with the standard GPT-2 causal language-modeling objective using AdamW and a batch size of 256. For each data scale, we held out 10% of the data for validation and used validation loss for early stopping and checkpoint selection. To assess the effect of input size, we trained models on three amounts of caregiver language: 0.1M, 1M, and 10M words. Each condition was run with three different random seeds.

Reward models

For each of the four caregiver feedback dimensions above, we train a reward model that maps a child utterance to the corresponding valence of the caregiver response, as annotated using the procedures described in Appendix 3.

All reward models share the same pre-trained model: `microsoft/deberta-v3-xsmall` [He, 2022], which we fine-tune using linear regression layer and a mean squared error (m.s.e.) loss to predict the target reward for a given utterance.

All reward models are trained on the CHILDES child–caregiver conversation pairs (one row per child utterance with the immediately following caregiver response), the same pre-processed pipeline above. We split the data into 90% training and 10% held-out evaluation with a fixed random seed. We used the AdamW optimizer with an initial learning rate of 1.41×10^{-5}, a batch size of 128, and early stopping based on MSE on the held-out validation set. The best checkpoint is selected on held-out MSE and used as the frozen reward model during PPO fine-tuning.

Topline Reward Model

The topline reward model was obtained using the same procedure, except that instead of learning the mapping between child utterances and the feedback valence, it was trained on the controlled minimal-pair datasets used in Zorro and BLiMP. More specifically, the model was trained to classify sentences from these datasets according to their labels (i.e., grammatical or ungrammatical).

Appendix 5: RL Fine-tuning Details

After pretraining on child-directed CHILDES utterances, we fine-tune each language model with Proximal Policy Optimization [PPO; Schulman, 2017], implemented using Hugging Face's TRL library. For each fine-tuning step, we (a) sample utterances from the current language model, (b) compute the corresponding rewards using the reward model, and (c) update the model weights using PPO.

Hyper-parameter	**Value**
Max PPO total steps	6000
Batch size	1024
Mini-batch size	512
KL controller	adaptive, target KL* = 6
Entropy coef. c_e	1×10^{-3}
LM mixing coef. λ_{LM}	1×10^{-3}
Score clipping	disabled
Sampling	T=1.0, top-p=1.0, top-k=0
Precision	bf16 mixed
Eval frequency	every 100 steps
Log frequency	every 10 steps
Early stopping	no reward gain for 10 log windows

TABLE 4.4 – PPO fine-tuning hyper-parameters, shared across the four caregiver-feedback rewards, the grammar topline, and all pretraining scales.

To obtain a diverse set of generated utterances, we randomly prompted the model with short utterance prefixes (the first one or two tokens) sampled from the pretraining data, namely the caregiver utterances used to train the input-based baseline model. We additionally included an entropy regularization term (0.001) in the loss.

The models were fine-tuned for a maximum of 6,000 steps, and the best checkpoint was selected based on mean reward. To discourage excessively short or long utterances, we applied rejection sampling by assigning a reward of −1 to generated utterances that were shorter than three tokens or failed to produce an end-of-sequence token within 20 tokens.

To mitigate language drift, a known issue in Reinforcement Learning (RL) fine-tuning, we also added a small language-modeling regularization term (weight = 0.001). We used the TRL default optimizer and learning rate. Other Hyperparameters are shown in Table 4.4. All other PPO hyperparameters were not changed from the default values as implemented in the Huggingface TRL library.

Each PPO run uses one NVIDIA H100 GPU (96 GB) with bf16 mixed precision and completes in approximately 6 GPU-hours.

Conclusion et perspectives

5.1 Conclusions de l'étude

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5.2 Ouverture et perspectives

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Liste des publications

Articles à comité de lecture

Co Auteur 1, Co Auteur 2, Co Auteur 3. "*Un long titre d'article*". Un journal qui a accepté mon travail.
DOI: a1b2c3d4.xxxxx.99999

Articles en préparation

Co Auteur 1, Co Auteur 2, Co Auteur 3. "*Le long titre de mon second article*". Le journal de mes rêves.

Conférences internationales (premier auteur)

Co Auteur 1, Co Auteur 2, Co Auteur 3, Co Auteur 4. "*Titre de la présentation*". Conférence de Sciences, Quelque part, 2020.

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RÉSUMÉ

Suspendisse vitae elit. Aliquam arcu neque, ornare in, ullamcorper quis, commodo eu, libero. Fusce sagittis erat at erat tristique mollis. Maecenas sapien libero, molestie et, lobortis in, sodales eget, dui. Morbi ultrices rutrum lorem. Nam elementum ullamcorper leo. Morbi dui. Aliquam sagittis. Nunc placerat. Pellentesque tristique sodales est. Maecenas imperdiet lacinia velit. Cras non urna. Morbi eros pede, suscipit ac, varius vel, egestas non, eros. Praesent malesuada, diam id pretium elementum, eros sem dictum tortor, vel consectetuer odio sem sed wisi.

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MOTS CLÉS

mot clé 1, mot clé 2, mot clé 3, mot clé 4

ABSTRACT

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KEYWORDS

keyword 1, keyword 2, keyword 3, keyword 4